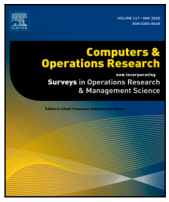
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Collaborative optimization of routing and storage strategy of multi-period multimodal transport in an uncertain environment

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ABSTRACT

This study uses a storage service strategy to investigate a multiperiod multimodal transport routing and transit planning problem (MPMT-RTP). First, a mathematical integer programming model is used to describe the problem in a deterministic environment. Second, two robust optimization models are established to address the uncertainty of the freight rate fluctuations in each period and the storage space in the total set, thereby improving the solving efficiency of the model through a robust equation transformation. Subsequently, hybrid heuristics are proposed to solve the above problems, and a series of small-scale experiments are conducted to verify the accuracy of the model and effectiveness of the algorithm. Finally, this study conducts case experiments based on survey data. The results show that the total operating cost is reduced by 13.31 % via strategic multiperiod multimodal transport planning. A price fluctuation study shows that under different price fluctuations, the transportation and storage periods of some orders change accordingly, and under three specific price fluctuations, the operating costs are 1085100, 967330, and 948770, respectively. Therefore, the results of this study can provide theoretical guidance for enterprises to optimize their resource allocation paradigms and better adapt to freight rate fluctuations to minimize their operating costs.

1. Introduction

With the rapid development of economic globalization and regional economic integration, trade between countries and regions has become increasingly frequent (Jing and Yang, 2022). In recent years, the proportion of multimodal transport modes in the total commodities volume, such as international railroad intermodal transport (China-Europe Railway Express) and road-river intermodal transport, has continued to increase (He et al., 2021; Chen et al., 2023). Multimodal transport is not a simple superposition of segmented transport. Its efficient operation requires multiple transport modes, transit nodes, logistics service providers, and users to work together under a predefined service strategy. The freight rates of different transport modes is the primary consideration for enterprises, and high or low freight rates directly affect the choice of transport modes and routes. In addition, factors such as market demand, commodities characteristics and transport routes, can lead to a shortage of commodities capacity on some transport sections, where commodities demand is concentrated. And this will result in additional time and costs for multimodal transport services (Zhen et al., 2019). From the perspective of a logistics service provider, factors such as transport time, cost and commodities characteristics can be taken into account, in order to develop a low-cost transport strategy. In this way, resource allocation and transport networks can be optimized, transport efficiency can be improved and costs can be reduced. In practice, service providers can store en-route and co-ordinate differences in transport capacity and supply and demand, to reduce logistics costs holistically. The example in Fig. 1 explains the effect of collaborative optimization of the routing and storage strategy of multi-period multimodal transport. The basic scenario does not involve a storage policy, and the shipment of an order can only occur during one period. However, some modes of transport during specified periods may not meet demand, and alternative modes may be too expensive. It is more efficient for the operator to advance the shipment time, as shown in Fig. 1. If the quantity of goods required for the third period is 40, the transit and storage costs are 5 at a time. For simplicity, it is assumed that highway, railway, waterway, and airplane costs take the same value in different paths, that is, 50, 30, 10, and 150, respectively. The transportation cost was 190, and the sum of the transportation, transit, and storage costs was 200 (including two transit charges). In the extended scenario with storage strategies, in the second period, the goods arrive at nodes 2 and 3 by highway transport successively and are then stored at node 3 for a period. In the third period, goods arrive at node 4 via railway transport. The transportation and comprehensive operating costs were 130 and 140, respectively. Reasonably advanced transportation and storage strategies led to a 26% and 30%

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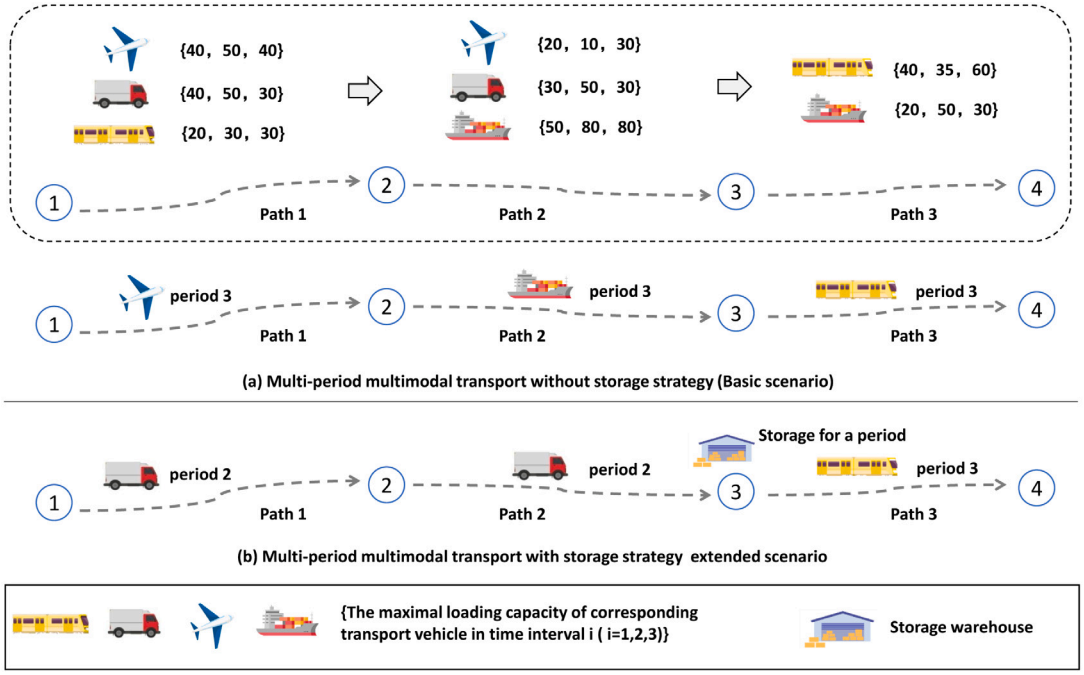


Fig. 1. Schematic diagram of MPMT-RTP.

decrease in transportation and operating costs, respectively. This study aimed to examine the following questions. (1) How to implement an organic combination of multimodal transport modes to achieve the rational use of resources based on differences in transport prices and capacity. (2) Dividing multimodal transport time into multiple periods, where building warehouses and storing commodities, can better enable the transport, consolidation, and transshipment of commodities and improve transport efficiency. (3) How can multi-period multimodal transportation routes and storage strategies be optimized to reduce integrated operating costs?

In order to address the above issues, this study introduces a new multi-period multimodal transport routes and storage strategy optimization problem (MPMT-RTP), which takes multimodal transport research further. The objective of the model is to simultaneously determine the multimodal transport strategy and the storage strategy in the warehouse for each transport period. In this paper, an integer planning model is used to build the MPMT-RTP model, where the transport price for each period and the value of storage space in the warehouse for each order of commodities are determined. Instead of the traditional multimodal route optimization problem, the new problem considers the balance between transit warehouse locations, multimodal transportation costs, and commodity storage costs.

In addition, multimodal transport operating practices face a variety of uncertainties that can have a significant impact on their operating costs. Common methods to address data uncertainty in optimization problems include sensitivity analysis, chance-constrained models, stochastic programming, and robust optimization. A sensitivity analysis is suitable for addressing deterministic problems or parameter variations within a certain range. It is difficult to deal with structural uncertainty in sensitivity analysis; that is, problems with uncertainty in the model structure itself. Chance-constrained models usually require precise estimates of the probability distributions of uncertain parameters and, therefore, rely heavily on large sample sizes. This study's transport price and storage space parameters can be measured accurately. However, this is both time and resource-intensive. In addition, the measurement process may be inaccurate, or specific data may be deliberately hidden. Thus, obtaining accurate distribution information is difficult and expensive. Similarly, stochastic programming requires the probability distributions of uncertain parameters to be explicitly known or estimated. In addition, stochastic programming is more sensitive to randomness, which means that different random samples may lead to different optimization results for the same set of parameters. This renders the stability of the results challenging. Unlike chance-constrained models and stochastic programming, robust optimization does not require prior knowledge of the specific probability distribution of uncertainty; instead, it considers all scenarios within a certain range of uncertainty. Robust optimization ensures that a feasible solution can be obtained despite uncertainty and helps prevent the failure of a decision-making solution owing to unknown uncertainty. In addition, robust optimization is usually computationally simpler than methods considering detailed probability distributions. This avoids the mathematical treatment of complex distributions; thus, it is computationally more efficient.

Therefore, this study further builds on the deterministic model by using robust optimization as the main modeling strategy, to address the uncertainty in storage space requirements and freight rates. In this study, the uncertainty in freight rates may arise from natural conditions, unexpected conditions, international influences, and other factors. And, multimodal transport orders involve numerous types of transport commodities, which are not uniform in size; therefore, there may be multiple package sizes for the same size of commodity, which leads to uncertainty in the storage space required for the commodities. The main contributions of this study are as follows: First, this study introduces a new multi-period multimodal transport routes and storage strategy optimization problem. Second, this study discusses the problem of co-optimization of multi-period multimodal transport routes and storage strategies under two scenarios: freight rate uncertainty and storage space uncertainty. The robust optimization method was combined with a multimodal network design to improve the feasibility of a multimodal operation plan under data uncertainty, with robust equation transformation to enhance its solving efficiency. Subsequently, the classical k-shortest path algorithm, the iterative greedy algorithm (IG), and the adaptive large neighborhood search algorithm (ALNS) were improved for the proposed robust random optimization model, resulting in a hybrid heuristic algorithm called MKIGALNS. MKIGALNS helps establish an integrated solution for operational decision making under the cross-border multimodal transport mode, which is based on cross-time transportation and in-transit storage. Finally, we conducted

numerical experiments to provide useful insights for multimodal operation managers and to promote modeling theory, robust methods, and optimization algorithms from theory to practice.

The rest of the paper is organized as follows. In Section 2, the relevant literature is reviewed. Section3 discusses the deterministic models (MPMT-RTP). Section 4 presents robust models and their pairwise transformations for the two cases of uncertain storage space and uncertain freight rates. Section 5 presents the MKIGALNS hybrid heuristic algorithm and MGA algorithm. Section 6 describes the numerical experimental setup based on the classical Sioux-Falls, followed by an analysis of the results of the relevant arithmetic cases and the performance of the MKIGALNS algorithm. Section 7 presents a related case study in the context of realistic data. Section 8 presents the relevant managerial implications of this study. Finally, Section 9 discusses conclusions and directions for future research.

2. Literature review

2.1. Logistics, transport routes and the locations of facilities

We first refer readers to three survey papers by Wen et al. (2019), Liang et al. (2021), Xie et al. (2022) that provide reviews of the importance of trade route selection and infrastructure construction for cross-border e-commerce transportation. All show that for efficient e-commerce logistics, the rational choice of trade routes and infrastructure optimization is essential. Moreover, to reduce logistics costs, many scholars have applied appropriate mathematical modeling and optimization methods to solve practical problems more accurately. Abudureheman and Nilupaer (2021) used an optimized cross-border ecommerce logistics distribution model, to search for the optimal route with the shortest total journey distance that satisfies environmental requirements, using an efficient search algorithm. Based on these papers, we mainly focus on a review of logistics transport routes and facility locations. Wang et al. (2021a) developed a bi-objective mathematical model considering the siting of logistics facilities and two-stage vehicle path optimization, they solved it using a two-stage hybrid algorithm to achieve maximum resource utilization over multiple service cycles. Shaabani et al. (2023) based on the special characteristics of the port, established the corresponding mathematical model, to solve the multi-product maritime transport path problem. To address the practical distribution problems faced by third-party logistics providers, Kramer et al. (2019), Sun et al. (2020), Wang et al. (2021b), and Niu et al. (2021) have planned logistics transport routes and modes for different constraints and purposes. Hence, these cannot be applied to some special transport problems. Additionally, specific constraints must be included in certain special transport problems. For the transportation of certain hazardous commodity, Wu et al. (2022) considered the location of the facility and various potential risks to its transport network design.Guo et al. (2021a) proposed an integer-planning model for splitting transport by order, to reduce the adverse impact of the transport environment on commodity, in response to the consolidation limit of the transported commodity. Furthermore, Dukkanci et al. (2019) investigated the green hub site problem to determine the optimal locations of hubs to satisfy route demand, and plan vehicle allocation and routing to minimize total emissions. However, above studies did not consider the relationship between multi-temporal transport and storage of goods in transit. In response to that element, Wang et al. (2023), Agra et al. (2022), Reddy et al. (2022), Darmawan et al. (2021), Wang et al. (2021c) have studied the logistics route problem for multiple time periods, and have involved warehouses to achieve operational strategies and facility location optimization. These emphasize the relevance of the rational planning of logistics routes for warehouse operations at different times of the year. These studies provide comprehensive and deep insights covering multiple aspects of cross-border e-commerce transport, ranging from route selection to facility siting and multi-timed transport. Recently, the transport path planning and infrastructure selection problems have received extensive attention in both academia and practice; however, very few studies have focused on the coordinated optimization of paths and storage strategies under multimodal transport settings, especially multimodal transport problems in uncertain environments.

2.2. Multimodal transport

In addition to the problems of logistics transport routes and facility locations, another major concern for this study is multimodal transport. Numerous scholars have studied multimodal transport network designs from different perspectives. Basallo-Triana et al. (2021) analyzed the literature on multimodal transport network design from the perspective of hub locations. Ye et al. (2021), Real et al. (2021), and Dong et al. (2020) have proposed planning models to provide better decision support for designing multimodal transport network systems. For the high carbon emission problem, Luo et al. (2021), Heinold and Meisel (2020) have addressed cost, emission, and service objectives, by rerouting the optimal path of freight orders through multimodal networks . Other notable studies focus on the optimization of multimodal transport routes and service strategies. Guo et al. (2021b) studied the location strategy for emergency relief facilities in multimodal networks, and developed a collaborative coverage model to improve the resilience of the network. Ghaderi and Burdett (2019) studied transport networks with location-route problems, and developed a two-stage stochastic planning model for the decision problem of hazardous materials to minimize transport costs and risks. Furthermore, Li et al. (2022) have considered bottleneck congestion and service capacity constraints at key multimodal node facilities to build a multimodal multi-cargo freight network equilibrium model to predict and analyze the spatial distribution of container freight flows. Zweers and van der Mei (2022) provided a model for balancing multimodal transport in transport costs and stochastic transport times owing to delays and overbooking, and developed a heuristic method to find the optimal multimodal transport path. These studies provide insights into different aspects of multimodal transport network design, such as system optimization, emergency response, and environmental friendliness, and provide useful insights for further research in this area. Although many scholars have conducted extensive research on designing multimodal transport networks, effective models that consider the trade-offs between the storage, transport and transit costs incurred in multiple time periods are still lacking. When addressing the above problems, it is difficult for the existing optimization algorithms to ensure their efficiency.

2.3. Uncertain environments

The current literature on the problems of logistics transport routes and facility locations for multimodal transportation has paid less attention to potential uncertainty environments than to deterministic environments. In multimodal transport practice, fluctuations in demand, price, and time are affected by policies, natural conditions, and other unforeseen situations. In the context of uncertainty environments, some scholars have conducted exploratory studies that combine logistics transport strategies with robust optimization. Liu and Li (2021) considered uncertainties in the volume and composition of raw materials for the location-route planning problem of a recycling facility, and proposed a finite adaptive distribution robust optimization paradigm to reduce unobservability and ambiguity in recycling planning. Moreover, Maiyar and Thakkar (2020) developed a mixed-integer nonlinear robust optimization model to develop near-optimal shipment volume, route selection, and center location decisions, to facilitate the transportation of a single type of food commodity. Mohammadi et al. (2021) considered the uncertainty of multimodal transport demand, transport time, and unloading time, and developed a stochastic model for multimodal transport by introducing a risk metric to minimize the total expected logistics distribution costs. In addition, to address the uncertainties that

exist in the reality of post-disaster situations, Sun et al. (2021), Balcik and Yanıkoğlu (2020), Yang et al. (2023), Zhang et al. (2023) have used robust optimization methods to achieve a fast response and minimize total casualties and total system cost, considering the uncertainties that exist in reality. Similarly, in solving the vehicle route problem with time windows, robust optimization methods have also been addressed for the presence of uncertainties, to minimize the delay risk associated with the time window constraint and to achieve cost minimization (Zhang et al., 2022; Wu and Hifi, 2020; Liu et al., 2023a). Furthermore, robust optimization methods and models have also been used to minimize the cost and risk of total supply chain links, in response to uncertainty factors in supply chain design and transport in different areas (Tarei et al., 2022; Golsefidi and Jokar, 2020; Guan et al., 2022). Those demonstrate the diversity of research approaches adopted by scholars to address the challenges of uncertainty to improve the robustness of transport and supply chain systems. This series of studies provides targeted and effective solutions for dealing with complex and changing uncertain environments and contributes important ideas and methods for achieving efficient and sustainable logistics and transport systems. The above analysis show that transport problems involving uncertain environments, including those exhibiting temporal uncertainty (Zhang et al., 2022; Balcik and Yanıkoğlu, 2020; Mohammadi et al., 2021) and demand uncertainty (Liu et al., 2023a; Sun et al., 2021; Guan et al., 2022), have been more fully studied in the literature. The existing studies concerning multimodal transport and path optimization problems under uncertain environments are still deficient, and they have not yet fully considered the uncertainty factors that exist in actual multimodal transport scenarios. For example, storage space uncertainty is caused by the diversity of the types of transported goods, and transport price uncertainty is caused by environmental or policy changes. Table 1 presents a comparison of the main references.

Previous studies have focused on transport route planning and infrastructure selection, considering their importance in different areas. For example, in sustainable development it helps to reduce the environmental impact and promote green transport, while in supply chain management it helps to optimize the logistics process and increase the resilience and effectiveness of the supply chain. The synergistic operation of multiple transport modes, including highways, railways, and waterways, can improve the efficiency of the overall transport network. However, several operational and planning issues still need to be addressed around synergistic multimodal transport operations, particularly in uncertain environments. This paper, however, introduces a new problem involving the co-optimization of multitemporal and intermodal routing and storage strategies, incorporating transhipment costs, storage costs and intermodal transport costs into the study. The impact of in-transit storage on intermodal operating costs and the degree to which storage decisions affect the flexibility of transport mode choices are described, providing new ideas and methods for further developments in the field of intermodal transport. Subsequently, this paper carries out an in-depth study on the co-optimization problem under the uncertainties of tariffs and storage space and applies the robust polyhedral uncertainty set-based optimization method, which provides a new perspective for improving the robustness of logistics and transportation and reducing risk. Under this model, enterprises can adjust the sizes of their uncertainty budgets to more comprehensively and accurately assess the impacts of different price scenarios or storage spaces for different commodity types on multimodal transport systems and formulate more robust multimodal transport strategies. This can help enterprises better adapt to fluctuations in shipping prices and transported commodities, reduce the cost of transport, and achieve improved competitiveness, which is important for solving real transport problems. This is of great significance for solving the uncertainty factors in actual transport scenarios. In this paper, we design an efficient hybrid heuristic algorithm, MKIGALNS, and verify that it has excellent accuracy and speed and achieves good performance in complex multitemporal multimodal transport cases, including storage decision-making and path planning. The proposed approach provides theoretical and algorithmic support for the realization of optimized multimodal transport strategies and solves the capacity constraints, transport price uncertainty, and storage space uncertainty of commodities.

3. Problem formulation

3.1. Problem description

During the startup period, the operator selects the required transit warehouse from a range of candidate warehouses, to provide storage for commodity. By coordinating transportation routes and storage strategies while also meeting the transportation needs of all orders, the operator reduces the total operating costs. The conditions that need to be satisfied for this problem can be briefly described as follows: (1) Delivery of commodities from the node of origin to the node of destination, (2) Existence of multiple modes of transportation between different nodes of the transportation network. There are differences in transportation distances, freight rates, and transportation capacities between the two nodes by different modes of transportation, (3) Transit costs are incurred when different modes of transportation are transferred at the node, (4) commodities may be taken to transit warehouses for storage during transit, and storage fees are payable according to the storage volume, and (5) The commodities can be transported to the terminal warehouse in advance, and inter-time transportation is allowed.

3.2. Parameter and variable definitions

The model in this study is defined on a directed graph, $G = \{\Phi, A\}$, where Φ denotes the set of all vertices in the network, which consists of a set of multimodal transit points (transit silos) M, virtual source nodes s^q, and virtual sink nodes b^q together, $\Phi = M \bigcup \{s^q\} \bigcup \{r^q\}$. For all arcs $(i,j) \in A : \{(i,j) i,j \in \Phi, i \neq j\}$, the distance is related to the modes of transport and is denoted as d_{ij}^n.$N = \{n_1, n_1, \ldots, n_z\}$ denotes the set of transport modes. A transport vehicle can transit through the modes of transport at each candidate point M. The transit cost is denoted by β_n^m. x_q denotes the customer demand for each transport order, where Q denotes the set of orders, and $q \in Q$ denotes the customer demand x_q for each order to arrive in time K. Due to certain factors in multimodal transport, the capacity constraint v_{ij}^{kn} and the cost per unit of transport c_{nk} in each arc change over time, and therefore, the whole transport time K is divided into multiple periods. $K = \{k_1, k_1, \ldots, k_m\}$ denotes the set of transport times. The decision maker selects several nodes to construct warehouses at the candidate node set M to ensure that the transportation process is smooth and the cost is minimized. Meanwhile, integrate the impact of freight rate fluctuations and capacity constraints of different modes of transportation at different periods and locations. Store orders that cannot be transported at a certain period because of capacity constraints and store orders that are costly to transport at a certain period because of freight rate increases. And μ_i denotes the cost of building a warehouse at candidate point i. I_i^k denotes the maximum available area of transit warehouse i in period k. ℓ_i denotes the unit storage cost of the transit warehouse. η_k^q denotes the storage space in the transit warehouse for the order commodities transported in a certain period.

The following variables are used to describe the model. The binary decision variable ψ_i indicates whether the decision maker establishes a warehouse at candidate point i. If a warehouse is established, $\psi_i = 1$; otherwise, $\psi_i = 0$. The binary decision variable h_i^{qk}indicates whether order q is stored at the transit node i in period k. If yes, $h_i^{qk} = 1$; otherwise, $h_i^{qk} = 0$. The binary decision variable z_{ij}^{qkn}indicates whether order q arrives at transit node j from transit node i using transport mode n in period k. If yes, $z_{ij}^{qkn} = 1$; otherwise, $z_{ij}^{qkn} = 0$. The binary decision variable y_{nm}^{qki} indicates whether order q changes from transport mode n to transport mode m at transit node i in period k; If yes, $y_{nm}^{qki} = 1$; otherwise, $y_{nm}^{qki} = 0$.

Table 1
Characteristics of the problems involved in some relevant studies.

Literature	MT	MP	Transit costs	SS	Uncertainties	TCC/SRC	RO
Guo et al. (2021a)						✓	
Luo et al. (2021)	✓		✓				
Liu et al. (2021)				✓	✓		✓
Mohammadi et al. (2021)	✓				✓		
Ye et al. (2021)	✓					✓	
Reddy et al. (2022)		✓		✓			
Wu et al. (2022)		✓		✓			
Liu et al. (2023b)					✓		✓
Shaabani et al. (2023)				✓		✓	
Wang et al. (2023)		✓				✓	
this study	✓	✓	✓	✓	✓	✓	✓

MT : Multimodal transport MP : Multiple periods SS : Storage Strategy.
TCC/SRC : Transport capacity/storage resource constraints RO : Robust Optimization.

3.3. Nominal model with strategic transportation decision

Based on the above conditions, assume that the freight rate c_{nk} and the storage space of the commodity η_k^q are deterministic values. The mathematical model is as follows:(NM-strategic)

$$\begin{aligned}\min \quad & \sum_{q\in Q}\sum_{n\in N}\sum_{k\in K}\sum_{\{j|(i,j)\in\widehat{A}\}} x_q d_{ij}^n c_{nk} z_{ij}^{qkn} \\ & + \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in\widehat{A}\}}\sum_{k\in K} x_q \beta_n^m y_{nm}^{qkj} \\ & + \sum_{i\in M}\psi_i\mu_i + \sum_{k\in K}\sum_{q\in Q}\sum_{i\in M} x_q \ell^i h_i^{qk}\end{aligned} \tag{1}$$

subject to :

$$\sum_{n\in N} z_{ij}^{qkn} \le 1 \qquad \forall k\in K,(i,j)\in\widehat{A},q\in Q \tag{2}$$

$$\sum_{n,m\in N} y_{nm}^{qki} \le 1 \qquad \forall i\in M,k\in K,q\in Q \tag{3}$$

$$\begin{aligned}& z_{ij}^{qkn}+z_{jl}^{qkm} \le 1 + y_{nm}^{qkj} \\ & \forall(i,j)\in\widehat{A},(j,l)\in\widehat{A},j\in M,\{n,m\}\in N,k\in K,q\in Q\end{aligned} \tag{4}$$

$$\begin{cases} \sum_{j\in M}\sum_{n\in N}\sum_{k\in K} z_{ij}^{qkn} - \sum_{j\in M}\sum_{n\in N}\sum_{k\in K} z_{ji}^{qkn} = 1, i = s^q; \\ \sum_{j\in M}\sum_{n\in N}\sum_{k\in K} z_{ij}^{qkn} - \sum_{j\in M}\sum_{n\in N}\sum_{k\in K} z_{ji}^{qkn} = -1, i = b^q; \\ \sum_{j\in M}\sum_{n\in N}\sum_{k\in K} z_{ij}^{qkn} - \sum_{j\in M}\sum_{n\in N}\sum_{k\in K} z_{ji}^{qkn} = 0, otherwise \end{cases} \quad \forall i\in M,q\in Q,(i,j)\in\widehat{A},(j,i)\in\widehat{A} \tag{5}$$

$$\sum_{q\in Q} x_q z_{ij}^{qkn} \le \upsilon_{ij}^{kn} \qquad \forall(i,j)\in\widehat{A},k\in K,n\in N \tag{6}$$

$$\sum_{q\in Q} h_i^{qk} x_q \eta_k^q \le I_i^k \psi_i \qquad \forall i\in M,k\in K \tag{7}$$

$$\begin{aligned}& \sum_{\{i|(l,i)\in\widehat{A}\}}\sum_{n\in N} z_{li}^{qkn} + h_i^{qk-1} - h_i^{qk} \\ & = \sum_{\{j(i,j)\in\widehat{A}\}}\sum_{n\in N} z_{ij}^{qkn} \forall i\in M,k,k-1\in Kq\in Q\end{aligned} \tag{8}$$

$$\psi_i, z_{ij}^{qkn}, y_{nm}^{qki}, h_i^{qk} \in \{0,1\} \tag{9}$$

The objective function (1) means minimizing the sum of transportation cost, transit cost, construction cost and storage cost. Constraint (2) means that at most one mode of transportation can be used between transportation nodes i and j in period k. Constraint (3) means that the transit operation can only be performed at the node. Constraint (4) is a logical constraint, expressing the correspondence of transportation modes both before and after the transit operation. Constraint (5) means that the node flows are balanced. Constraint (6) presents a capacity constraint. Constraint (7) means that the amount of commodities stored cannot exceed the capacity constraint of the transit warehouse. Constraint (8) means the balance of incoming and outgoing transit commodities volumes. Finally, constraint (9) means indicates parameter and variable characteristics.

The relaxation of the model is a modified version of the resource-constrained shortest path problem, which determines the minimum cost path from a source node to a sink node and satisfies a set of constraints (Irnich and Desaulniers, 2005). Prior studies have demonstrated that this problem is at least as difficult as an NP-complete problem (Handler and Zang, 1980). Because the multi-period multimodal transport routing and transit planning problem is a generalization of the above problem, it is also NP-complete.

4. Robust optimization

4.1. Uncertainty in storage space

In this section, the robustness problem under storage-space uncertainty is discussed. When designing intermodal transportation and warehousing strategies, service providers must meet freight demand even in worst-case scenarios; otherwise, user concerns about whether orders will be delivered on schedule will limit service provider profitability. The central theme of robust optimization is therefore to choose a set of uncertainties such that the resulting solution is not affected by the uncertainties. Common robust uncertainty sets include Box Uncertainty Set, Ellipsoidal Uncertainty Set and Polyhedral Uncertainty Set. When modeling parameter uncertainty with a Box Uncertainty Set, assuming that this uncertainty applies to all parameter values may lead to overly conservative results. Compared to the box uncertainty set, the ellipsoid uncertainty set is more flexible and allows the main axis direction and length of the ellipsoid to be adjusted according to the real situation to capture the uncertainty characteristics of the system more accurately; however, this increases the complexity of the problem. In the case of polyhedral uncertainty, the proportional change in the uncertain parameters cannot exceed a certain threshold value (called the uncertainty budget) rather than preventing deviations in all parameters. Decision-makers can balance robustness and performance by adjusting the uncertainty budget to meet actual needs. Linear programming techniques and tools can still be used to solve polyhedral uncertainty, making it relatively easy to solve. This section applies the uncertainty budget method to construct the uncertainty set of storage space. The uncertainty set F consists of confidence intervals for the storage space requirements of the commodities in each order. Service

providers can adjust multimodal transport service levels by varying uncertainty budgets. Because storage space requirement only appears in the constraint, the robust model is described as follows.(RUH1) subject to : (2)–(6),(8)–(9)

$$\max_{\tilde{\eta}\in F}\left\{\sum_{q\in Q} h_i^{qk} x_q \eta_k^q\right\} \le I_i^k \psi^i \qquad \forall i\in M, k\in K \tag{10}$$

Constraint (10) ensures that the amount of commodities stored cannot exceed the capacity constraint of the warehouse, even in the worst-case scenario.

It is assumed that $\tilde{\eta}_k^q$ is unknown, approximating the nominal value $\overline{\eta}_k^q$ on the symmetry interval $U_k^q = [\overline{\eta}_k^q - \hat{\eta}_k^q, \overline{\eta}_k^q + \hat{\eta}_k^q]$, where $\hat{\eta}_k^q$ is the largest deviation from the nominal value. Let $U_k = U_k^1 \times \cdots \times U_k^{|Q|}$ and $U = U_1 \times \cdots \times U_{|K|}$. For storage space $\tilde{\eta}_k^q$, introduce a level of uncertainty $\xi_k^q \in [0,1]$, indicating the proportional deviation of storage space from its nominal value. Thus, for each q and k, the storage space $\tilde{\eta}_k^q$ lies in the interval $[\overline{\eta}_k^q - \xi_k^q\hat{\eta}_k^q, \overline{\eta}_k^q + \xi_k^q\hat{\eta}_k^q]$. In practical decision making, $\overline{\eta}_k^q$ is obtained by the decision maker based on past data. The uncertainty level ξ_k^q measures the estimated accuracy of the overall prediction. The uncertainty budget $\Gamma_k \in [0, |Q|]$, where $|Q|$ is the number of orders, which ensures that there are at most $\lfloor\Gamma_k\rfloor$ orders in which the storage space changes according to $\hat{\eta}_k^q$ and there is one order in which the storage space changes according to $(\Gamma_k - \lfloor\Gamma_k\rfloor)\hat{\eta}_k^q$.

$$\begin{aligned} F = \{&(\tilde{\eta}_k^q) | \tilde{\eta}_k^q \in [\overline{\eta}_k^q - \xi_k^q\hat{\eta}_k^q, \overline{\eta}_k^q + \xi_k^q\hat{\eta}_k^q], \forall k\in K, q\in Q, \\ &\sum_{q\in Q}\xi_k^q \le \Gamma_k,\ \forall k\in K, 0\le\xi_k^q\le 1, \forall k\in K, q\in Q\} \end{aligned} \tag{11}$$

In the uncertainty set, for all $k \in K$, if $\Gamma_k = 0$, the feasible element in the uncertainty set is $\overline{\eta}_k^q$, and thus, the problem comes down to the nominal issue, where there is no protection against uncertainty. For all $k \in K$, if $\Gamma_k = |Q|$, each uncertain storage space takes its worst-case value, which yields a conservative solution. If $0 < \Gamma_k < |Q|$, for some $k \in K$, the uncertainty budget allows the decision maker to balance the degree of constraint protection and the level of conservativeness.

The following theorem reformulates the robust counterpart as integer linear programming.

Theorem 1. *Given the polyhedral uncertainty set $U = U_1 \times \cdots \times U_{|K|}$ and uncertainty budget Γ_k, $k \in K$. The nonlinear robust model (RUH1) is equivalent to the following equivalent mixed-integer linear programming problem (RUH2), where the corresponding dual variables w_i^k, p_{qk}^i is used to linearize the model(RUH2) (Bertsimas and Sim, 2004b).*

subject to: (2)–(6),(8)–(9)

$$\sum_{q\in Q}\overline{\eta}_k^q h_i^{qk} x_q + \Gamma_k w_i^k + \sum_{q\in Q} p_{qk}^i \le I_i^k \psi^i \qquad \forall i\in M, k\in K \tag{12}$$

$$p_{qk}^i + w_i^k \ge \hat{\eta}_k^q h_i^{qk} x_q \qquad \forall q\in Q \tag{13}$$

$$p_{qk}^i \ge 0 \qquad \forall q\in Q \tag{14}$$

$$w_i^k \ge 0 \tag{15}$$

We consider the nonlinear robust counterpart of constraint (7) in the problem (Bertsimas and Sim, 2004a)

$$\sum_{q\in Q}\overline{\eta}_k^q h_i^{qk} x_q + g(x,\Gamma_k) \le I_i^k \psi_i \qquad \forall i\in M, k\in K \tag{16}$$

Of which

$$\begin{aligned} g(x,\Gamma_k) = &\max_{\{S_k\bigcup\{t_k\}|S_k\subseteq F_k, |S_k|=\lfloor\Gamma_k\rfloor, t_k\in F_k\setminus S_k\}} \\ &\times\left\{\sum_{q\in S_k}\hat{\eta}_k^q h_i^{qk} x_q + (\Gamma_k - \lfloor\Gamma_k\rfloor)\hat{\eta}_k^{t_k} h_i^{t_k k} x_{t_k}\right\} \end{aligned} \tag{17}$$

The storage space subset selection process $\{S_k \bigcup\{t_k\} | S_k \subseteq F_k, |S_k| = \lfloor\Gamma_k\rfloor, t_k \in F_k\setminus S_k\}$, and the corresponding function $\sum_{q\in S_k}\hat{\eta}_k^q h_i^{qk} x_q + (\Gamma_k - \lfloor\Gamma_k\rfloor)\hat{\eta}_k^{t_k} h_i^{t_k k} x_{t_k}$ are equivalent to solving the following linear optimization problem.

$$g(x,\Gamma_k) = \max_{\{w_k\}}\left\{\sum_{q\in Q}\hat{\eta}_k^q h_i^{qk} x_q \xi_k^q\right\} \tag{18}$$

subject to

$$\sum_{q\in Q}\xi_k^q \le \Gamma_k\ \forall q\in Q, k\in K \tag{19}$$

$$0\le\xi_k^q\le 1 \qquad \forall k\in K, q\in Q \tag{20}$$

Because the value of the optimal solution of constraint (18) consists of $\lfloor\Gamma_k\rfloor$ as 1, and a variable as $\Gamma_k - \lfloor\Gamma_k\rfloor$, the optimal values of function (17) and function (18) are equal, and we obtain the uncertainty set defined in (11).

Defining w_i^k and p_{qk}^i as the dual variables of (19) and (20), respectively, the dual of the above is

$$\min_{\{w_i^k, p_{qk}^i\}} \Gamma_k w_i^k + \sum_{q\in Q} p_{qk}^i \tag{21}$$

subject to (13)–(15)

Because of strong duality, for all $\Gamma_k \in [0, |Q|]$, because constraint (18) is feasible and bounded, the dual problem (21) is also feasible and bounded, and their objective values are the same. Substituting objective function (21) into the constraint (10), we can obtain that constraint (10) is equivalent to the constraints (12) and (13) in the linear optimization problem (RUH2). The introduced uncertain budget approach ensures that the solution to problem (RUH2) will remain feasible with high probability if the storage space $\tilde{\eta}_k^q$ has $\lfloor\Gamma_k\rfloor$ orders varying in the bounded interval and the rate of change on a path varies with $(\Gamma_k - \lfloor\Gamma_k\rfloor)\hat{\eta}_k^q$.

In this section, we use the approach in line with Bertsimas and Sim (2004a) to flexibly adjust the level of conservatism of robust solutions in terms of the probabilistic bounds of constraint violations. The uncertainty parameters are known as the lower and upper bounds and are assumed to obey a symmetric probability distribution. If the storage space $\tilde{\eta}_k^q$ reaches $\lfloor\Gamma_k\rfloor$ orders vary within a certain range, and the storage space for one order varies from $(\Gamma_k - \lfloor\Gamma_k\rfloor)\hat{\eta}_k^q$, the solution to the problem remains feasible with high probability. Let $n = |Q|$, $v = (\Gamma_k + n)/2$, $\mu = v - \lfloor v\rfloor$, The a priori bounds of the problem can be approximated as follows:

$$B(n,\Gamma_k) \le (1-\mu)C(n,\lfloor v\rfloor) + \sum_{l=\lfloor v\rfloor+1}^{n} C(n,l) \tag{22}$$

where

$$C(n,l) = \begin{cases} \frac{1}{2^n}\ if\ l=0\ or\ l=n \\ \sqrt{\frac{n}{2\pi(n-l)l}}\exp\left(n\log\frac{n}{2(n-l)} + l\log\frac{n-l}{l}\right)\ otherwise \end{cases} \tag{23}$$

4.2. Uncertainty in freight rate

In order to more accurately model the freight uncertainty of cross-border trade, this study adopts a robust optimization method with uncertain parameters having multi-range characteristics. Since potential prices depend on different transportation periods, including all possible transportation values in one range would produce a large range of forecasts and may lead to overly conservative results. Using extensive freight rate data, decision makers can determine nominal and interval forecasts of freight rates under different scenarios. Decompose the uncertainty space $\tilde{c}_{nk}$ of freight rates into multiple ranges of $r \in R$, whereby in period k, the nominal freight rate in range r is denoted by $\bar{c}_{nk}^r$, and the maximum deviation of the freight rate in range r from the nominal freight rate is denoted by $\hat{c}_{nk}^r$, $\hat{c}_{nk}^r = \varepsilon^n \bar{c}_{nk}^r$ and ε^n is the level of price disturbance for different transportation modes.

The 0–1 variable w^r_{nk} is used to indicate whether the freight rate $\tilde{c}_{nk}$ is in the range of r for different transportation modes in period k. Let $\tilde{c}_{nk} \in [\bar{c}^r_{nk} - \hat{c}^r_{nk}p^r_{nk}, \bar{c}^r_{nk} + \hat{c}^r_{nk}p^r_{nk}]$, where $p^r_{nk} \in [0,1]$ denotes the proportional deviation of the freight rate in period k from its nominal rate in range r. The uncertainty budget Γ_{rn} limits the number of freight rates that fall within the range r. The uncertainty budget Γ_n limits the number of deviations from the nominal freight rate within a given range. The current robust optimization problem can be represented as the following mixed-integer programming problem (RUP1).

$$\begin{aligned}\min\ & \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in\widehat{A}\}}\sum_{k\in K} x_q\beta^m_n y^{qkj}_{nm} \\ &+ \sum_{k\in K}\sum_{q\in Q}\sum_{i\in M} x_q h^{qk}_i \ell^i + \sum_{i\in M}\psi_i\mu_i \\ &+ \max_{\{\tilde{c},w\}} \sum_{k\in K}\sum_{n\in N}\tilde{c}_{nk}\sum_{q\in Q}\sum_{\{j|(i,j)\in\widehat{A}\}} x_q d^n_{ij} z^{qkn}_{ij}\end{aligned} \tag{24}$$

subject to: (2)–(9)

$$\tilde{c}_{nk} = \sum_{r\in R} c^r_{nk} \qquad \forall k\in K, n\in N \tag{25}$$

$$(\bar{c}^r_{nk} - \hat{c}^r_{nk}p^r_{nk})w^r_{nk} \le c^r_{nk} \le (\bar{c}^r_{nk} + \hat{c}^r_{nk}p^r_{nk})w^r_{nk} \quad \forall k\in K, n\in N, r\in R \tag{26}$$

$$\sum_{r\in R} w^r_{nk} = 1 \quad \forall k\in K, n\in N \tag{27}$$

$$\sum_{k\in K} w^r_{nk} \le \Gamma_{rn} \quad \forall r\in R, n\in N \tag{28}$$

$$\sum_{k\in K}\sum_{r\in R} p^r_{nk} \le \Gamma_n \quad \forall n\in N \tag{29}$$

$$w^r_{nk}\in\{0,1\}, 0\le p^r_{nk}\le 1 \quad \forall r\in R, n\in N, k\in K \tag{30}$$

Objective function (24) represents the total cost of the worst-case minimization, and constraints (25)–(27) ensure that the unit freight rate for transportation mode n is a value within r in each time period k. The number of freight rates within constraint (28) is limited. Constraint (29) denotes the price deviation of transport prices. Constraint (30) indicate parameter attributes.

Introduction: Considering the total proportional change in freight rates Γ_n and the maximum number of rates in each range Γ_{rn}, (RUP1) is equivalent to the following integer linear programming problem:

$$\begin{aligned}\min\ & \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in\widehat{A}\}}\sum_{k\in K} x_q\beta^m_n y^{qkj}_{nm} \\ &+ \sum_{k\in K}\sum_{q\in Q}\sum_{i\in M} x_q h^{qk}_i \ell^i + \sum_{i\in M}\psi_i\mu_i \\ &+ \max_{\{\tilde{c},w\}} \sum_{k\in K}\sum_{n\in N}\sum_{r\in R}(\bar{c}^r_{nk}w^r_{nk} + \hat{c}^r_{nk}u^r_{nk})\sum_{q\in Q}\sum_{\{j|(i,j)\in A\}} x_q d^n_{ij} z^{qkn}_{ij}\end{aligned} \tag{31}$$

subject to : (2)–(9), (27)–(28)

$$u^r_{nk} \le w^r_{nk} \quad \forall r\in R, n\in N, k\in K \tag{32}$$

$$\sum_{r\in R}\sum_{k\in K} u^r_{nk} \le \Gamma_n \quad \forall n\in N \tag{33}$$

$$w^r_{nk}\in\{0,1\}, u^r_{nk}\ge 0 \quad \forall r\in R, n\in N, k\in K \tag{34}$$

Proof. In solving the RUP1 problem, the objective function (24) is a maximization problem. Since the coefficients of the decision variables and the objective function are non-negative, in order to find the optimal uncertain tariff, the following conditions need to be satisfied: $c^r_{nk} = (\bar{c}^r_{nk}+\hat{c}^r_{nk}p^r_{nk})w^r_{nk}$ and $\tilde{c}_{nk} = \sum_{r\in R}(\bar{c}^r_{nk}+\hat{c}^r_{nk}p^r_{nk})w^r_{nk}$. Let $u^r_{nk} = p^r_{nk}w^r_{nk}$, then $c^r_{nk} = \bar{c}^r_{nk}w^r_{nk} + \hat{c}^r_{nk}u^r_{nk}$, where $0\le u^r_{nk}\le w^r_{nk}$. For $\forall r\in R, n\in N, k\in K$,in some case, if $u^r_{nk} = 0$, then $p^r_{nk} > 0$ is a suboptimal choice in that case (Düzgün and Thiele, 2010). Substitute c^r_{nk} and u^r_{nk} into (RUP1) to obtain the worst-case mixed-integer programming problem.

Introduction: The robust equivalence problem under freight uncertainty is equivalent to the following problem (RUP2)

$$\begin{aligned}\min\ & \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in\widehat{A}\}}\sum_{k\in K} x_q\beta^m_n y^{qkj}_{nm} \\ &+ \sum_{k\in K}\sum_{q\in Q}\sum_{i\in M} x_q h^{qk}_i \ell^i + \sum_{i\in M}\psi_i\mu_i \\ &+ \sum_{k\in K}\sum_{n\in N}\zeta_{kn} + \sum_{k\in K}\sum_{n\in N}\sum_{r\in R}\varphi^r_{kn} + \sum_{r\in R}\sum_{n\in N}\Gamma_{rn}\gamma^r_n + \sum_{n\in N}\Gamma_n\gamma_n\end{aligned} \tag{35}$$

subject to : (2)–(9)

$$\pi^r_{nk} + \gamma_n \ge \hat{c}^r_{nk}\sum_{q\in Q}\sum_{\{j|(i,j)\in\widehat{A}\}} x_q d^n_{ij} z^{qkn}_{ij} \qquad \forall k\in K, r\in Rn\in N \tag{36}$$

$$-\pi^r_{nk} + \zeta_{kn} + \gamma^r_n + \varphi^r_{kn} \ge \bar{c}^r_{nk}\sum_{q\in Q}\sum_{\{j|(i,j)\in\widehat{A}\}} x_q d^n_{ij} z^{qkn}_{ij} \qquad \forall k\in K, r\in R, n\in N \tag{37}$$

$$\pi^r_{nk}, \gamma^r_n, \gamma_n, \varphi^r_{kn} \ge 0, \zeta_{kn}\in\mathbb{R} \quad \forall k\in K, n\in N, r\in R \tag{38}$$

Proof. The optimal solution of linear relaxation constraints (27)–(28)and (32)–(34) is equal to the corresponding integer optimal solution. Let $\zeta_{nk}, \gamma^r_n, \pi^r_{nk}, \gamma_n$ be the pairwise variables of constraints (27)–(28) and (32)–(33). φ^r_{kn} is the dual variable of the linear relaxation constraint in (34), i.e., $0 \le w^r_{nk} \le 1$. The optimal objective value of the original problem and the optimal objective value of the pairwise problem in (35) coincide. Constraints (36) and (37) are the pairwise constraints for variables u^r_{nk} and w^r_{nk}, respectively. Constraint (38) indicate parameter attributes.

5. Algorithm design

This section deals with two algorithms, of which the MKIGALNS algorithm is presented in Sections 5.1–5.3, and the MGA algorithm is briefly described in Section 5.4. Among them, the hybrid heuristic algorithm MKIGALNS is obtained by improving the classical k-shortest path algorithm, the iterative greedy algorithm (IG) and the adaptive large neighborhood search algorithm (ALNS) for the proposed robust stochastic optimization model. (1) Incorporating multimodal weighted transportation cost conversion rules in the generation of the initial set of routes, combined with the characteristics of the research problem. (2) Constructing a route feasibility fitness function in conjunction with the IG framework. Finally, (3) Designing an efficient adaptive mechanism in the ALNS link to improve the algorithm's solving efficiency, and adding removal operators and insertion operators, to improve the global search performance of the algorithm. A brief introduction of MKIGALNS follows Fig. 2. And the key points of the algorithm design are as follows:

5.1. Transport network and initial route acquisition(mksp, modified K-shortest path algorithm)

5.1.1. Code rules

The research questions in this study involve multiple periods as well as multiple modes of transportation, requiring the following coding structure: KN—X, where"K" indicates the period, "N" indicates the transportation mode (1–4 indicates road, rail, water, and air transport respectively), and "X" indicates the node number. For example, "21-08—21-20" means that the commodity is delivered from node 08 to node 20 by road transport in the second period. "21-08—22-08" means that the commodity is converted from road transport to rail transport at node 08 in the second period.

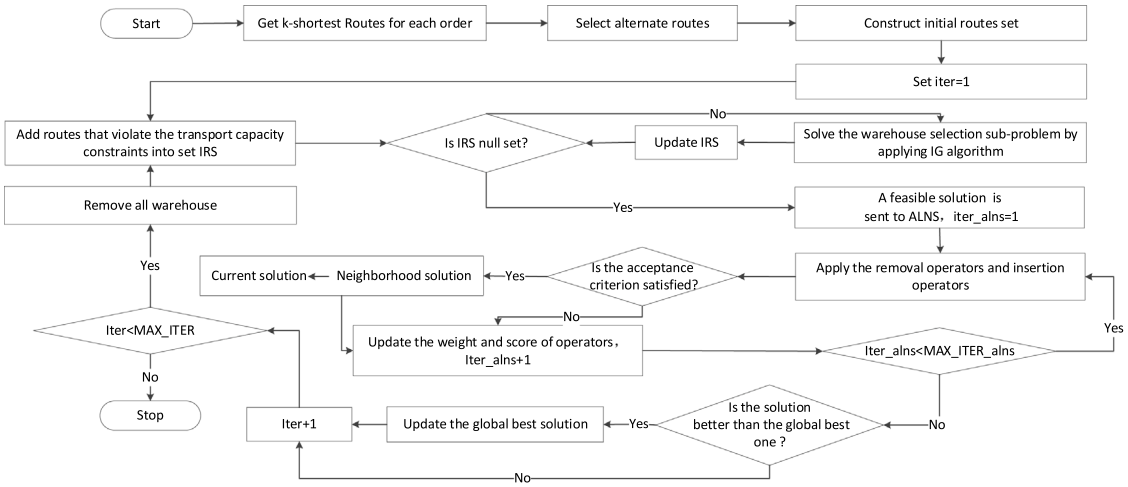


Fig. 2. Flowchart of MKIGALNS.

5.1.2. Network preprocessing and transport route initialization

Three types of connections may occur between two nodes of a multimodal transport network. (1) Without any line segment for connection, i.e., the commodity cannot go straight from one node to another node. (2) A straight line connects the two nodes, i.e., the commodity can be transported from one node directly to the other node by the mode of transport, represented by the line. (3) There are several different line segments connecting two nodes, i.e., commodity can be transported directly from one node to another node through a variety of transportation modes, represented by the line.

To ensure that the network parameters conform to the arithmetic form of the MKSP algorithm, the freight rate differences of different transport modes for different route sections, were equivalently converted into the transport distance differences of the corresponding transport modes in the following form: $\alpha' = \varepsilon\alpha/\varepsilon'$, where α and α' denote the actual and converted distances, respectively, and ε and ε' denote the unit freight rate and their converted values, respectively.

The k-shortest path algorithm was used to calculate the top N shortest routes for each order in each period and to store them in a set of alternative routes BX. Subsequently, the shortest route for each order in each time period was used as the initial route CS.

Note that, for 5.1.2, the network must be transformed into a multilayer network. For example, if there are 6 original nodes, a distance using numbers 1–6 indicates road transportation distance, and 7–12 indicates air transportation distance,13–18 indicates water transportation distance. If the difference between two nodes is 6, the distance is zero (effectively, the same node).

5.2. Transit warehouse service decision based on iterative greedy algorithm

This section further considers factors, such as the current route and transportation mode capacity constraints, based on the initial set of routes obtained in Section 5.1. To minimize operating costs, an iterative greedy algorithm (IG) is used to select the best-located transit warehouse and make it available for storage at the right period. This section is divided into the following steps: feasibility identification of routes, transit warehouse setting, and storage-object selection.

5.2.1. Feasibility identification of routes

In turn, check whether the initial set of routes in the first period can meet the capacity demand of each transport order, and if there is an infeasible route (i.e., the capacity of a mode of transportation is not sufficient to meet the transportation demand, in the current period of the route), then enter 5.2.2 for transit warehouse setting.

5.2.2. Transit warehouse setting

Several concepts related to the potential costs of commodity storage strategies are first described—overload section and transportation task completability.

(a) Overload section: Arc $i\widehat{j}$ in period k has the maximum transportation volume of v_{ij}^{kn} with transportation mode n, and the transport volume for order q is x_q. The difference between the two is $t_{ij}^{qnk} = v_{ij}^{kn} - x_q$. The road section is considered to be overloaded with the corresponding transport mode, if the following occurs $\{i\widehat{j}\,|t_{ij}^{qnk} < 0\}$.

(b) Transportation task completability: The impact of constructing a transit warehouse on the transportation tasks of all orders passing through the node was evaluated through transportation task completability. We defined the transportation task completability of order q in arc $i\widehat{j}$ through transport mode n as p_{ijqn}, then $p_{ijqn} = \min\{t_{ij}^{qnk}, 0\}$. The lowest value of the transportation task completability for order q among all sections of the current route is $p_{ijqn}^{*} = \min_{n\in N, i\widehat{j}\in\widehat{A}} p_{ijqn}$.

The choice of a transit warehouse is influenced by factors, such as transportation volume,freight rates differences between different periods, and transportation task completability. A storage strategy is selected by the IG under the roulette mechanism, based on the high or low potential cost of the transit warehouse at different locations along the route. Let the cost of setting up a transit warehouse at pair network node i and storing the commodity in that warehouse be c_{ijqn}, with the calculation as shown below:

The unit freight rate difference between transportation mode n in period interval k and its subsequent contiguous period intervals through arc $i\widehat{j}$, before and after the construction of a transit warehouse at node i as shown in l_{ijnk}. s_{qi} is a 0–1 variable indicating that if node i in the transportation plan of order q undergoes a transit, then $s_{qi} = 0$; otherwise, it is 1. I_{qik} indicates the maximum amount of leased space, that can be offered to the commodity of order q at the transit warehouse in period interval k located at node i.

$$c_{ijqn} = \beta_1\left(-g_{ijqn}\right) + \beta_2 l_{ijnk} + \beta_3 s_{qi} + \beta_4 e_{qik} \quad \forall q \in Q \tag{39}$$

$$\beta_1 + \beta_2 + \beta_3 + \beta_4 = 1 \quad \beta_1, \beta_2, \beta_3, \beta_4 \geq 0 \tag{40}$$

$$g_{ijqn} = \sum_{\{j|(i,j)\in\widehat{A}\}} \left(p''_{ijqn} - p'_{ijqn}\right)\left(\frac{1+\left|p*'_{ijqn}\right|}{1+\left|p*''_{ijqn}\right|}\right) \tag{41}$$

$$e_{qik} = M\left|\min\{I_{qik} - x_q, 0\}\right| \tag{42}$$

Constraint (41) indicates the improvement in order q's transportation task completability before and after adding a transit warehouse at

node i. p'_{ijqn} and p''_{ijqn} denotes the transportation task completability before and after the establishment of a transfer warehouse at node i,respectively. $p*'_{ijqn}$ and $p*''_{ijqn}$ denotes the minimum value of transportation task completability before and after the establishment of a transfer warehouse at node i,respectively. The first part of the right-hand side of constraint (41) indicates the different values of transportation task completability of order q in arc $i\widehat{j}$, before and after the addition of the transfer warehouse. The second part on the right-hand side uses the characteristics of $\left|p*'_{ijqn}\right| \geq \left|p*''_{ijqn}\right|$ in the cumulative value of transportation task completability, to amplify the first part of the different values. Constraint (42) indicates the penalty condition for unavailable warehouses, where M is an extreme value. If the storage space I_{qik} offered by the transit warehouse, added at node i for order q in period interval k, is not sufficient to meet the storage requirements x_q, i.e., $I_{qik} - x_q < 0$, then the storage solution is considered infeasible and a penalty is imposed. Conversely, if the transit warehouse added at node i provides sufficient storage space for order q during period interval k, both $\min\{I_{qik} - x_q, 0\}$ and e_{qik} are zero.

5.2.3. IG optimization process

This section describes the use of an iterative approach to gradually build transit warehouses in a service network, to ensure that each order is completed as required. To avoid falling into a local optimum, the IG expands the search range by introducing a random mechanism in the process of adding site locations and optimizing the site selection.

Step 1: Remove the transit warehouses contained in the current solution, temporarily deposit the orders that do not meet the current period capacity constraints into the infeasible routing set (IRS), and deposit the remaining orders into the set FRS.

Step 2: Statistic of all the overload sections in the current transportation plan, the starting nodes of all the overload sections are stored in the set B_{qk}. All the nodes in the set B_{qk} are the preselected locations for adding the transit warehouse.

Step 3: The nodes in the set B_{qk} are taken out in turn. Take node i in B_{qk} as an example, record the orders that are blocked at this node in the current period, due to capacity constraints. Next, the cost of building a transit warehouse at that node and storing the commodity of order q(i.e. c_{ijqn}) is calculated and recorded in J_{iqk}.

Step 4: The nodes included in B_{qk} are ranked in order of c_{ijqn} from highest to lowest, and node i is selected through $\left\lfloor \varepsilon_1^{\rho_1} \left| B_{qk} \right| \right\rfloor$, where ε_1 is a random number, $\varepsilon_1 \in [0,1]$, and for the sake of further enhancing randomness introduce $\rho_1 \geq 1$, this study assigns $\rho_1 = 5$. Next, node i is used as the preselected location for the transit warehouse.

Step 5: After determining node i, each order contained in J_{iqk} is ranked in the order of c_{ijqn} from highest to lowest, and the orders to be stored are selected by $\left\lfloor \varepsilon_2^{\rho_2} \left| J_{iqk} \right| \right\rfloor$, where ε_2 is a random number, $\varepsilon_2 \in [0,1]$, and $\rho_2 = 5$.

Step 6: Update the transportation task completability and move orders that satisfy the capacity constraints from IRS to FRS. Repeat Steps 2–6 until the IRS is an empty set.

Step 7: Repeat the above process and check backwards in turn the next period until all orders are cleared for transportation. If all alternative nodes in a route are setting transfer warehouses or store commodities multiple periods until the last period, the route is still infeasible. We then select the subsequent alternative route for the corresponding order from the alternative route set BX, and reexecute the steps described in Section 5.2.

5.3. Transport route optimization based on adaptive large neighborhood search

5.3.1. ALNS algorithmic framework

The adaptive large neighborhood search (ALNS), first proposed by Ropke and Pisinger (2006) , is an extension of the large neighborhood search (LNS) algorithm, and ALNS is a popular method for solving transportation planning problems with multiple transportation modes (Guo et al., 2022; Liu et al., 2023a; SteadieSeifi et al., 2017). The main idea of ALNS is to search for a large neighborhood by removing a given number of nodes from a solution and then reinserting them to construct a new solution. The removal and insertion operators are chosen adaptively at each iteration based on their historical success. The process of the ALNS algorithm for the multi-period multimodal transport route optimization problem in an uncertain environment is as follows:

Step 1: The route policy and storage policy of IG are used as the initial solution for s_0. Subsequently, s_0 is assigned to the current solution s and the optimal solution s^*. Initialize the operator weights and start the first iteration of the first search.

Step 2: Select the removal and insertion operators and assign the current solution s to the neighborhood solution s'.

Step 3: Call the removal operator to deal with s'.

Step 4: Call the insertion operator to deal with s'.

Step 5: If s' satisfies the acceptance criteria, s' is assigned to the current solution s.

Step 6: If $Z(s) < Z(s^*)$, then assign the current solution s to the optimal solution s^*, iteration number +1.

Repeat Steps 2–6 until the maximum number of iterations is reached. Then move to Step 7.

Step 7:If all the search rounds have been completed, output the optimal solution. Otherwise, update the operator scores and weights and return to Step 2 for the next search round.

5.3.2. Adaptive search mechanism

ALNS consists of a series of search rounds, with the weights being updated at the beginning of each search round based on the search results of the operator. Operator search results are counted by scoring. Let the initial score of the operator be 0. If the selected operator gets the global optimal solution, 50 points are added to the operator; if the selected operator gets the current optimal solution of the search round, 30 points are added to the operator; if the selected operator gets a feasible solution, 10 points are added to the operator. The initial value of the weight w_{i1} of operator i is 10 points, and the update process is as follows.

Assume that the number of uses and scores of operator i in search round j are ε_{ij} and π_{ij}, in turn. If operator i is selected in search round j, the weight of i in search round $(j+1)$ is $w_{i(j+1)} = (1-\theta)\, w_{ij} + \theta \pi_{ij} / \varepsilon_{ij}$, with θ as the equilibrium coefficient. Otherwise, the weight remains unchanged at w_{ij}. Each search round consists of ε iterations, and before the start of each iteration, ALNS selects the operators that will enter this iteration under a preset roulette mechanism. Assume that the weight of operator i at the j iteration is w_{ij}, then the probability of i being selected is $p_{ij} = w_{ij} / \sum_{h \in H} w_{hj}$, where H is the set to which the i operator belongs.

5.3.3. Penalty function and termination criterion

To further extend the search range appropriately, solutions that violate the transport capacity constraints are treated as penalties instead of deletions, i.e., this solution is retained and a penalty value is added to its objective function value.

$$\begin{aligned} \min & \sum_{q\in Q}\sum_{n\in N}\sum_{k\in K}\sum_{\{j|(i,j)\in \widehat{A}\}} x_q d_{ij}^n c_{nk} z_{ij}^{qkn} \\ & + \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in \widehat{A}\}}\sum_{k\in K} x_q \beta_n^m y_{nm}^{qkj} + \sum_{i\in M} \psi_i \mu_i \\ & + \sum_{k\in K}\sum_{q\in Q}\sum_{i\in M} x_q \ell^i h_i^{qk} - M \sum_{q\in Q}\sum_{n\in N}\sum_{\{j|(i,j)\in \widehat{A}\}} p_{ijqn} \end{aligned} \tag{43}$$

In this study, a termination criterion is adopted, which is similar to the simulated annealing-based algorithm proposed by Adulyasak et al. i.e., the neighborhood solution s' is retained with a probability

$e^{-(Z(s')-Z(s))/T}$, if the neighborhood solution s' is larger than the current solution s. The search process ends when the number of iterations reaches a given maximum.

5.3.4. Large neighborhood

In each round of iteration, the removal operator is called to destroy the transportation scheme of the current solution for the route n_p, which is a random number in the interval $[\varsigma_1|D|, \varsigma_2|D|]$. And $\varsigma_1, \varsigma_2 \in (0,1)$, $|D|$ are the number of routes involved in the current transportation scheme. For the selected route n_p, there are two ways to destroy the transportation scheme: deleting the transportation period and deleting the transportation mode.

5.3.5. Removal operators

Four general removal operators are used in the following section—Random removal (RaR), Basic worst removal (BWR), Related removal (ReR), and Request graph removal (RGR).

RaR operator: Randomly selects route n_p from the current solution and destroys its transportation scheme.

BWR operator: The cost change of route $i\widehat{j}$ before and after deletion is used as the deletion weight w_{ij} of route $i\widehat{j}$. The routes are ranked according to the weight of route deletion from highest to lowest, and according to $\left\lfloor \varepsilon_3^{\rho_3} |D| \right\rfloor$, the corresponding routes are selected to be temporarily stored in the set WS. ε_3 is a random number and $\varepsilon_3 \in [0,1]$, ρ_3 is a given parameter. In the BWR operator, reducing routes reduces transportation costs and affects transit costs. Therefore, the deletion weight is $w_{ij} = \alpha_1 \sum_{q\in Q}\sum_{n\in N}\sum_{k\in K}\sum_{\{j|(i,j)\in\widehat{A}\}} x_q d_{ij}^n c_{nk} z_{ij}^{qkn} + \alpha_2 \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in\widehat{A}\}}\sum_{k\in K} x_q \beta_n^m y_{nm}^{qkj}$, where α_1 and α_2 are the weighting parameters and $\alpha_1 + \alpha_2 = 1$.

ReR operator: A road section $i\widehat{j}$ is randomly deleted, and then some road sections with high similarity to road section $i\widehat{j}$ are selected and temporarily stored in set WS. The similarity between the road sections $j\widehat{l}$ and $i\widehat{j}$ is $Iden(ij,jl) = \alpha_3 d_{ij}^{jl} + \alpha_4 \left|q_{ij} - q_{jl}\right| + \alpha_5 \eta_{ij}^{jl}$, where α_3, α_4 and α_5 are the weight parameters and $\alpha_3 + \alpha_4 + \alpha_5 = 1$, and $\left|q_{ij} - q_{jl}\right|$ is the difference of transportation volume between section $j\widehat{l}$ and $i\widehat{j}$. If Use the same mode of transportation in sections $j\widehat{l}$ and $i\widehat{j}$, $d_{ij}^{jl} = 1$ else $d_{ij}^{jl} = 0$. If transporte the same order in sections $j\widehat{l}$ and $i\widehat{j}$, $\eta_{ij} = 1$ else $\eta_{ij} = 0$. Similar to BWR, network routes similar to route $i\widehat{j}$ are selected with a certain probability, which is determined by the parameters ε_4 and ρ_4.

RGR operator: The weights of sections $i\widehat{j}$ and $j\widehat{l}$ are the number of times sections $i\widehat{j}$ and $j\widehat{l}$ are used in the N_{best} optimal solutions. The higher the weight, the more similar sections $i\widehat{j}$ and $j\widehat{l}$ are assumed to be. The deletion procedure for the RGR operator is similar to that for ReR.

In addition to the above 4-type general removal operators, the following 4-type special removal operators were added to the characteristics of the $MPMT-RTP$ model.

AWR operator: The deletion weights of sections $i\widehat{j}$ in the BWR operator take into account the transportation costs and transit costs. Besides this, the deletion of section $i\widehat{j}$ will also have an impact on the transportation task completability of the remaining sections, and the deletion weight of the AWR operator is $w_{ij}^* = \lambda_1 w_{ij} + \lambda_2(\sum_{q\in Q}\sum_{n\in N} p'_{ijqn} - \sum_{q\in Q}\sum_{n\in N} p''_{ijqn})$, where, $\lambda_1 + \lambda_2 = 1$, p'_{ijqn} and p''_{ijqn} denote the transportation task completability before and after the deletion operation, respectively.

SBR operator: Randomly select transit warehouse j and delete all sections connected to j that have transportation tasks; this operation is repeated until the n_p sections are temporarily stored in a set of WS.

OBR operator: Select an order at random, delete all road sections of the order that are scheduled to be transported, and repeat this operation until the n_p road sections are temporarily stored in set of WS.

TPR operator: Select two nodes j and i at random in the network, and if nodes j and i are in the transport route of the same order, temporarily store all the road sections between the two nodes in the set WS.

5.3.6. Insertion operators

ALNS temporarily stores some transport sections in set WS by calling the removal operator, then destroy its transport scheme. The insertion operator is then called to recomplete the transport scheme in the set WS where the road section has been destroyed, until the set WS is empty.

For those that require additional periods to be numbered, start with the period of the previous section, and move backwards. For example, if the previous road section is transported in the second period, the current period can be considered for the subsequent periods starting from the second period. (The last period of the subsequent section of the current section, cannot exceed the final period.)For the road sections which the transport mode number has been deleted, the restoration process needs to take into full consideration factors, such as freight rate of the transport mode, and whether the nodes at both ends of the restored road section occur transit. For road sections where transportation periods are deleted, the transportation period of the repaired section shall not be earlier than its previous section, and the period of the order completed transportation shall not be later than the last period of the problem limit. The flow diagram is shown in Fig. 3. And based on the numbering format, we set the following rules.

Rule 1: The first position indicates the period, taking 1101——1102——2202——2214 as an example, the first A of the code on the left side of "——" is less than or equal to the first B of the code on the right side, (B-A)*C = total storage cost, where the difference between A and B indicates how many periods are stored, and C indicates the storage cost for each period.

Rule 2: The second position indicates the transportation mode. If the second position D of the left code is not equal to the second position E of the right code, transit cost will be paid.

Rule 3: Ensure that the last 3 positions of the left-hand side code and the last 3 positions of the right-hand side code have connecting lines in the network.

In this study, two general insertion operators (BGI and BRkI) and two improved insertion operators (AGI and ARkI) were designed to characterize the MPMT-RTP model.

BGI operator: The change in cost before and after the resettlement of road section $i\widehat{j}$ is used as the resettlement weight $\Delta Z_{ij}^{nk} = Z_{ij}^{nk} - Z_{(ij-)}^{nk}$ of road section $i\widehat{j}$. Select the lowest point of ΔZ_{ij}^{nk} as the new transportation scheme for road section $i\widehat{j}$, and $Z_{ij}^{nk} = \alpha_1 \sum_{q\in Q}\sum_{n\in N}\sum_{k\in K}\sum_{\{j|(i,j)\in\widehat{A}\}} x_q d_{ij}^n c_{nk} z_{ij}^{qkn} + \alpha_2 \sum_{j\in M}\sum_{q\in Q}\sum_{n,m\in N}\sum_{\{j|(j,l)\in\widehat{A}\}}\sum_{k\in K} x_q \beta_n^m y_{nm}^{qkj}$. Transportation scheme for road section $i\widehat{j}$ is $i\widehat{j}^* = \arg\min_{k\in K}\left\{\min_{n\in N}\Delta Z_{ij}^{nk}\right\}$.

BRkI operator: ΔZ_{ij}^{nk1} denotes the minimum placement weights resulting from the transportation reprogramming of road section $i\widehat{j}$. ΔZ_{ij}^{nkf} denotes the placement weight of the ranked j, resulting from the transportation reprogramming of road section $i\widehat{j}$. Unlike the BGI operator that always selects the placement weight at the minimum, the BRkI operator selects the placement location based on the regret value generated by the formulated transportation scheme. The transport scheme for road section $i\widehat{j}$ are $i\widehat{j}^* = \arg\max_{u\in U}\left\{\sum_{f=2}^{u}\left(\Delta Z_{ij}^{nkf} - \Delta Z_{ij}^{nk1}\right)\right\}$. In this study, $u = 2$ and $u = 3$ are used, and the corresponding operators are BR2I and BR3I.

The BGI and BRkI operators do not consider the effect of transportation scheme reformulation on transportation task completability. Define the improved placement weights as $\Delta Z'^{nk}_{ij} = \lambda_3 * \Delta Z_{ij}^{nk} + \lambda_4(\sum_{q\in Q}\sum_{n\in N} p'_{ijqn} - \sum_{q\in Q}\sum_{n\in N} p''_{ijqn})$,where $\lambda_3 + \lambda_4 = 1$, p'_{ijqn} and p''_{ijqn} denote transportation task completability before and after the removal operation, respectively.

An improved placement operator, AGI, was obtained by using $\Delta Z'^{nk}_{ij}$ replacing the BGI operator placement weight $\Delta Z_{ij}^{nk} = \Delta Z'^{nk}_{ij}$. Similarly, the improved placement operators AR2I and AR3I were obtained.

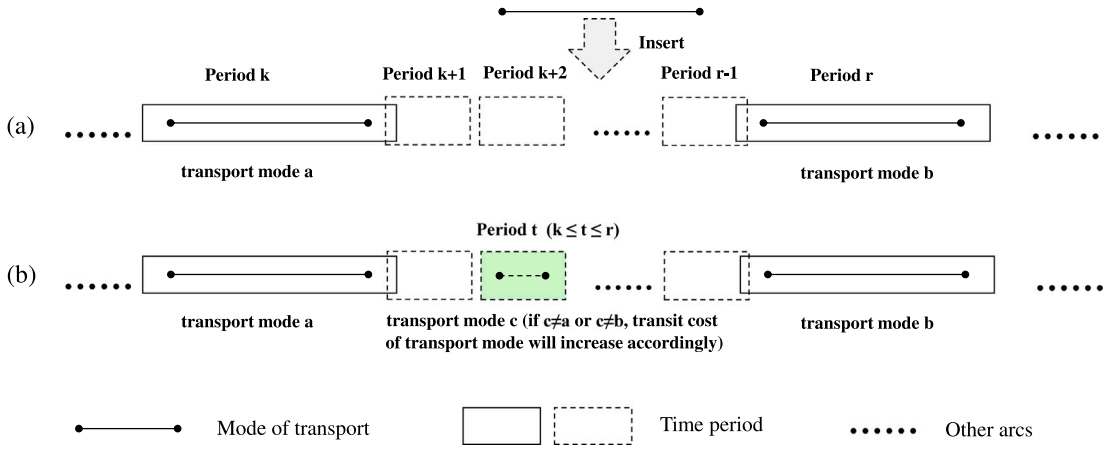


Fig. 3. Flow diagram of insertion operators.

Table 2
Schematic diagram of the coding.

first line	2	3	3	3	1	1	2	2	2
second line	1	1	1	2	2	2	2	3	3

5.4. The modified genetic algorithm (MGA)

Genetic algorithms are commonly used for multimodal transport optimization. According to the characteristics of the problem and genetic algorithm coding, this subsection adopts two-line natural number coding as the expression of the transport-storage strategy. The specific expression is as follows: the number in the first line of each gene represents the transport mode selected between adjacent nodes, and the number in the second line represents the transport period of the path. For example, in the first line of a gene, 1 indicates rail transport, 2 indicates road transport, 3 indicates water transport, and 4 indicates air transport. Table 2 shows a specific coding example of a chromosomal individual with 10 nodes (including the origin and destination) and 9 paths. The second mode of transport (road transport) is selected between nodes 1 and 2, and the third (water transport) is selected between nodes 2 and 3, etc. In the second line of the gene code, the value represents the period through the current path. In the example above, storage for a period occurs at nodes 4 and 8. Taking the Nominal model as an example, constraints (2)–(5) and (8)–(9) were considered in the matching solution of individual chromosome coding. Constraints (6) and (7) are considered in the fitness function.

Considering the large search space of the algorithm, the method of randomly generating the initial population was adopted to increase the individual diversity of the initial population while considering constraints (2)–(5) and (8)–(9) to avoid producing infeasible solutions. The fitness function of individual genes was constructed based on the objective functions (1) and (6)–(7). The smaller the value of the objective function, the larger the fitness value of the chromosome individuals. Simultaneously, individual fitness values are converted through rank-based fitness assignments, and individuals are ranked and selected to avoid early convergence owing to excessive individual fitness values in the initial operation of the genetic algorithm.

6. Experiments and results analysis

In this section, the experimental computer was configured with an Intel(R) E5-2678V3, 48G, 1050TI 4G, and optimization software CPLEX12.8.

6.1. Parameter setting

The path network is derived from the Sioux-Falls network, a network diagram often used in transportation network problems (Fan et al., 2022; Guo et al., 2020; Liu and Li, 2021; Ye et al., 2021). The transportation path and related parameters were adjusted through pre-experiments. The adjustment process was as follows:

Step 1: The Sioux-Falls network contained 24 nodes, 76 links, and 528 OD pairs. In this section of the experiment, 24 nodes of the Sioux-Falls network were used as the starting point, endpoint, and transit nodes of different multimodal transport orders. ($I = 24$)

Step 2: This section of the experiment uses four modes of transportation ($N = 4$): waterway, railway, highway, and airplane; that is, multiple modes of transportation can be selected for the same route. Therefore, after conducting the pre-experiments, we used The Sioux-Falls network as a basis and established the experimental route shown in Fig. 4, in which different line segments represent different transportation methods. If there is a line–segment connection between two nodes, then this mode of transportation exists on the road segment.

Step 3: In reality, transportation prices fluctuate to a certain extent because of special factors such as policy changes or international influences. Because this study adopts a multi-interval robust optimization method, this section assumes that there are three situations of transportation prices: falling, normal, and rising prices. Therefore, through relevant pre-experiments in the example experiments in this section, the nominal transportation prices of different transportation modes in different price periods are listed in Table 4, and the values of the other parameters are listed in Tables 3 and 5.

6.2. Parameter tuning

For many algorithms, the choice of parameters is critical to the performance. Appropriately tuning the parameters of an algorithm can significantly improve its performance. Many parameter tuning methods exist, such as the F-race method, Relevance Estimation, Value Calibration method, and ParamILS (Montero et al., 2014). Among these feasible methods, in this paper, we choose the ParamILS method for tuning and selecting the parameters (Hutter et al., 2007). An advantage of ParamILS is that it can be applied to a wide range of algorithms and problems with wider applicability. In addition, it enhances the user's ability to search the parameter search space and enables the user to obtain an appropriate configuration for solving the problem. Compared with the efficiency of ParamILS, F-racing can be excessively time-consuming for some problems.

ParamILS is a method for automatically tuning the parameters of an algorithm based on default parameter configurations, which optimizes the algorithm performance by employing an iterative search over the

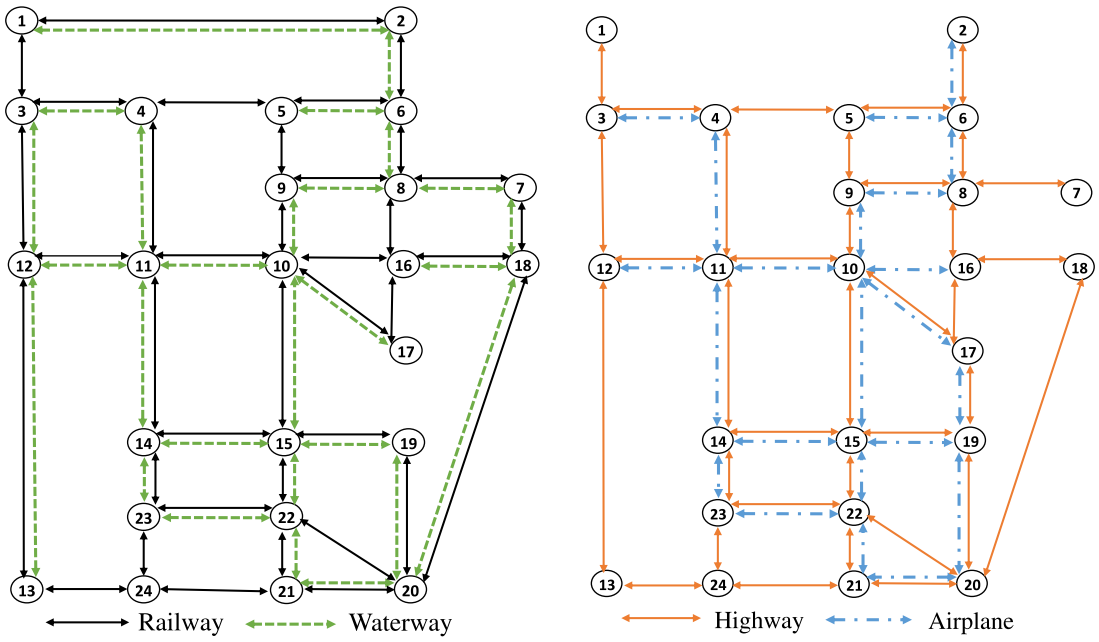


Fig. 4. The routes of multimodal transport.

Table 3
Parameters and their values.

Symbols	Parameter values	Symbols	Parameter values
Number of nodes $\|i\|$	24	Transit warehouse area $\|I_i^0\|$	1000
Orders $\|Q\|$	112	Storage costs $\|\mathscr{C}^i\|$	2.5
Mode of transport $\|N\|$	4	Number of time $\|T\|$	5

Table 4
Nominal freight rate setting for different rate ranges.

freight rate(Yuan/ton)	Highway	Railway	Waterway	Airplane
Price reduction period	0.125	0.025	0.010	0.180
Price normal period	0.200	0.050	0.020	0.360
Price increase period	0.325	0.075	0.030	0.540

Table 5
Transit cost setting.

Transit cost (Yuan/ton)	Highway	Railway	Waterway	Airplane
Highway	0	4	7	8
Railway	4	0	7	9
Waterway	7	7	0	9
Airplane	8	9	9	0

parameter space. Algorithm 1 outlines the operations of ParamILS. In each iteration round, it selects the current optimal parameter configuration and performs a local search in its vicinity. If the new parameter configuration outperforms the current optimal configuration in terms of performance, ParamILS updates the optimal parameters. This process is presented in Algorithm 2.

The processing factors in the ParamILS method correspond to the algorithm parameters. Table 6 presents the suggested levels for each processing factor. Table 7 shows the values of the optimization parameters of the algorithm. From the results of several preliminary tests, the maximum number of iterations was set as follows: if $|M| \leq 50$, $ITER^{MKIGALNS} = 2000$; if $50 < |M| \leq 100$, $ITER^{MKIGALNS} = 3000$; otherwise, $ITER^{MKIGALNS} = 4000$.

6.3. ALNS configurations

To select the best combination of operators to improve the efficiency of the ALNS process, two metrics were set for evaluating the operators' performance: the number of times the operators were used and the weights of the operators. Therefore, the operator selection criterion adopted in this section is as follows. In at least half of the instances, the operators are better than the average in both evaluation metrics. In this section, the operators' performances are compared, the number of invocations by each operator is recorded, and these operators are evaluated accordingly. Because the ALNS algorithm contains random parameters, 20 experiments were conducted for each arithmetic case, and the average value was taken as the final result. Tables 8 and 9 list the experimental results for the removal and insertion operators, respectively.

In Table 8, column 1 represents the evaluation metrics, where "Call" denotes the number of times the operator was selected for use, and "W" denotes the average of the final weights. Column 2 denotes a specific arithmetic example, columns 3–10 denote the experimental results for all operators, and column 11 denotes the average performance of the operators. The numbers that are better than the average for each instance are in bold. The definitions of the columns in Table 9 are the same as those in Table 8. Based on the above selection criteria, operators such as BWR, ReR, and BR2I met the requirements and were used in subsequent experiments (the selected operators are represented in bold in Tables 8 and 9).

Algorithm 1: Framework of ParamILS

$C_0 \leftarrow default\ parameter\ vector$
for $i \leftarrow 1\ to\ R$ **do**
  $C \leftarrow random\ parameter\ vector$
  if $better(C, C_0)$ **then**
    $C_0 \leftarrow C$
  end if
end for
$C_{ils} \leftarrow IterativeFirstImprovement(C_0, N)$
while *not terminationcriterion*() **do**
  $C \leftarrow C_{ils}$
  for $i \leftarrow 1\ to\ s$ **do**
    $C \leftarrow random\ parameter\ vector\ in\ N(C)$
    $C \leftarrow IterativeFirstImprovement(C, N)$
  end for
  if $better(C, C_{ils})$ **then**
    $C_{ils} \leftarrow C$
  end if
  if $p_{restart}$ **then**
    $C_{ils} \leftarrow random\ parameter\ vector$
  end if
end while
  return $overallbestC$

Algorithm 2: ParamILS $IterativeFirstImprovement$ Procedure

repeat
$C' \leftarrow C$
for each $C'' \in N(C')\ in\ randomized\ order$ **do**
  if $better(C'', C')$ **then**
    $C \leftarrow C''$
    break
  end if
end for
until $C' = C$
  return C

Table 6
Search space for parameters.

Parameter	*Values*	*Set size*
α_1	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
α_2	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
α_3	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
α_4	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
α_5	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
ς_1	{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40}	8
ς_2	{0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50}	8
λ_1	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
λ_2	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
λ_3	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10
λ_4	{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}	10

6.4. Comparison between CPLEX and algorithm

Based on the parameter setting and algorithmic configuration, the proposed algorithms, namely MKIGALNS and the modified GA, were tested on different instances. This section compares these heuristics with the exact algorithm in the CPLEX Solver for small instances.

As shown in Tables 10, 10 groups of differently sized cases were randomly selected from the network diagram to ensure the generality of the experiments, and the number of orders included in each group varied from 40 to 112. Column 2 presents the number of orders included in the example, and columns 3 and 4 present the optimal solution and computing time (s), respectively, calculated using CPLEX. Columns 5–7 present the results by MKIGALNS: the best solution and computing time (s), and the gap in percentage. Columns 8–10 present the results of MGA. Columns 7 and 10 present the relative differences between the optimal solution of CPLEX and the best solution of the heuristics.

The results in Table 10 show that the computing times of MKIGALNS and MGA are smaller than those of CPLEX, and the relative difference between the best solution of the two heuristics and the optimal solution of CPLEX is relatively small r, especially for MKIGALNS, which performs even better than MGA. These results demonstrated the effectiveness of the MKIGALNS algorithm.

6.5. Analysis of calculation examples

6.5.1. Analysis of the impact of storage strategy on operating costs

In this section, by setting up a control experimental group, the experimental results of the control group are compared and analyzed with the multi-modal route and storage co-optimization strategy proposed in this study(i.e. experimental group), under the same parameter values. The control group was not allowed to store commodities during transportation. The experimental results are shown in Table 11, where column 2 presents the number of orders included in the example. Columns 3 and 4 present the optimal solution and solution time (s) for the control group, respectively. Columns 5 and 6 present the optimal solution and solution time (s) for the experimental group, respectively. Column 7 presents the relative difference between the optimal solution of the control group and the experimental group, Gap = (optimal solution of the experimental group - optimal solution of the control group)/optimal solution of the control group. According to Table 11, in the ten test cases, the average optimal operating costs of the control group were greater than those of the experimental group, with an average difference of 7198. Figs. 5(a) and 5(b) illustrate the effects of storage strategies on transportation and transshipment costs, respectively. The solid lines represent the costs of the control group, which were consistently higher than the dashed lines representing the costs of the experimental group in all 10 cases. Therefore, it is evident that strategically storing goods helps reduce the overall operating costs.

6.5.2. Analysis of the impact of transit costs on transportation strategies

This section analyzes the intrinsic link between transit costs and transportation strategies under different scenarios. Based on the two experimental groups in the previous section regarding the impact of storage strategies on operating costs, scenario 1 was set as the transshipment cost experimental group (with the same values as in the previous section). Scenarios 2 and 3 were the control groups (with transshipment costs being 0.5 times and 2 times of scenario 1, respectively). The experimental results are listed in Table 12, where the second column shows the optimal solution for the transshipment cost in the scenario 1 experimental group (allowing storage), the third column shows the optimal solution for the scenario 1 control group (not allowing storage), and the fourth column shows the relative difference between the two optimal solutions. Columns 5–7 and 8–10 correspond to Scenarios 2 and 3 results, respectively.

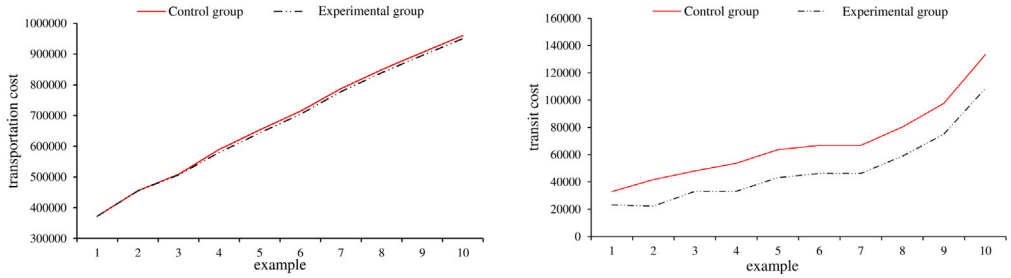
From the results of the eight experimental cases in Table 12, it can be observed that when storage is not allowed, with the increase in transshipment costs, the average operating cost of Scenario 3 control group is 710856.3, greater than the average optimal operating cost of Scenario 1 control group, which is 672788.8. The average increase in operating costs is 28067.5. As the transshipment costs decrease, the average optimal operating cost of the Scenario 2 control group is 609562.5, less than the average optimal operating cost of the Scenario 1 control group, with a decrease of 63226.3. Therefore, it is evident that trans-shipment costs have a corresponding impact on operating costs. Moreover, when storage was allowed, the operating costs of the scenario 3 experimental group were greater than those of the scenario 1 experimental group. However, the average increase in operating costs was 21565, which was less than the average increase in operating costs when storage was not allowed (28067.5). Hence, storage and transshipment coordinate with each other, jointly influencing operating costs.

6.5.3. Analysis of transport strategies under storage space uncertainty

This section analyzes the effect of the change of the robust parameter Γ_k on the multimodal transportation strategy when the storage

Table 7
Parameter setting.

α_1	α_2	α_3	α_4	α_5	ς_1	ς_2	λ_1	λ_2	λ_3	λ_4
0.7	0.3	0.5	0.4	0.1	0.2	0.4	0.7	0.3	0.8	0.2



(a) The impact of storage strategies on transportation costs (b) The impact of storage strategies on transit costs

Fig. 5. The impact of storage strategies on routes and transit plan.

Table 8
Performance comparison of removal operators for instances.

Param.	Instance	RaR	BWR	ReR	RGR	AWR	SBR	OBR	TPR	Avg.
Call	S-n40	151	**1650**	1255	611	**2298**	**1936**	**1488**	**2457**	1480.75
	S-n48	859	875	**1806**	1148	**2036**	**1921**	1515	**2353**	1564.13
	S-n56	351	**1757**	**1871**	1202	**2742**	**1719**	1380	**1639**	1582.63
	S-n64	1468	**1929**	**1755**	997	**2128**	1528	881	**1713**	1549.88
W.	S-n40	7.64	**12.06**	**10.61**	3.13	**12.01**	**8.97**	4.37	**9.44**	8.53
	S-n48	8.70	**11.30**	9.45	6.52	**12.77**	**12.42**	7.71	**11.38**	10.03
	S-n56	6.68	**10.78**	**13.52**	4.92	**13.49**	**12.46**	8.46	**11.73**	10.26
	S-n64	7.53	**12.31**	**12.95**	6.92	**11.73**	9.22	2.18	7.29	8.77

Table 9
Performance comparison of insertion operators for instances.

Param.	Instance	BGI	AGI	BR2I	BR3I	AR2I	AR3I	Avg.
Call	S-n40	1426	1739	**2867**	**2860**	**2434**	**3074**	2400
	S-n48	1058	1710	**2761**	**2634**	**3287**	**2950**	2400
	S-n56	1872	1686	2288	**2834**	**2994**	**2726**	2400
	S-n64	1901	1268	**2934**	**3215**	**2568**	**2514**	2400
W.	S-n40	9.88	7.03	**13.23**	**12.25**	9.94	**11.82**	10.69
	S-n48	5.03	6.01	**8.55**	**9.03**	**12.64**	**9.02**	8.38
	S-n56	5.33	8.38	9.35	**13.54**	**10.16**	**13.16**	9.99
	S-n64	9.02	8.22	**12.60**	11.00	**13.78**	**13.65**	11.38

space is uncertain. Taking the orders 93, 102, 106 as examples, we show the changes of transport period, storage period, transport volume and storage volume, when the parameter Γ_k in the robust model is changed. In Table 13, the transportation strategies are shown for three different Γ_k. Columns 1 to 5 of Table 13 show, in order, at Γ_k = 112, the transportation period, the transportation routes, the order numbers being shipped, the transportation mode, and the storage node when storage occurs.

The transportation strategy for order 106 in the table reveals that when Γ_k is 112, order 106 is transported at time k = 5 along route (24, 8). In route segments 24–21, order 106 is transported by rail, and in segments 21–8, it switches to waterway transportation. This indicates a change in transportation mode at transfer point 21 without cross-time transportation or storage activities. When Γ_k changed to 72, the transportation mode for order 106 remained the same as that when Γ_k was 112; however, the transportation time changed to k=3 without cross-time transportation or storage activities. However, when Γ_k changes to 40, there is a significant change in the transportation strategy for Order 106. First, at k = 3, order 106 is transported by rail in route segments 24–20. Subsequently, a storage decision is made to store the order for two periods in warehouse 20. Subsequently, at k = 5, order 106 switches to waterway transportation in route segments 20–8, along with transshipment, storage, and cross-time transportation activities. Additionally, the transportation traces of orders 93 and 102 in Table 13 show that, as Γ_k changes, they also undergo changes in their transportation strategies.

As can be observed from the data on the volume of goods transported by each mode of transport in each road section in the table, when Γ_k is 112 and 40, road sections (20–18) use the waterway to transport the volume of goods for the three orders at k=5. When Γ_k is 72, road section (20–18) uses waterway transport at k=3 to transport only the volume of goods of order 106, and at k=5, waterway transport is used to transport the volume of goods of orders 102 and 93. As seen from the transport volumes in the other sections of Table 13, they change as Γ_k changes. The data on the volume of goods stored in Warehouse 20 in the table show that when A is 112 and 72, the volume of goods for orders 102 and 93 are stored in Warehouse 20 at times k = 3 and k = 4, respectively. When Γ_k is 40, the volume of goods for order 106 is stored in Warehouse 20 in periods k = 3 and k = 4. Thus, as Γ_k changes, the amount of goods stored in the Warehouse also changes.

Therefore, when robust parameter Γ_k with uncertain of storage space changes, the period of transportation, the volume of commodities transported, and the volume of storage, all produce corresponding changes, to reduce the negative impact of transport commodities of different sizes on operating costs.

6.5.4. *Analysis of transportation strategies under freight rate uncertainty*

This section analyzes the effect of the change of the robust parameters (Γ_n, Γ_{rn}) on the multimodal transportation strategy when the freight rate is uncertain. The maximum deviation of the freight rate from its nominal value is assumed to be 20%, i.e., $\hat{c}^r_{nk} = 0.2\bar{c}^r_{nk}$. Taking the orders 3, 111, 112 as examples, we show the changes of transport period, storage period, transport volume and storage volume, when the combination of (Γ_n, Γ_{rn}) in the robust model is changed. In Table 14, the transportation strategies are shown for three different (Γ_n, Γ_{rn}) combinations. Columns 1 to 5 of Table 14 show, in order,at Γ_n = 2, Γ_{rn} = (0,1,4), the transportation period, the transportation routes, the order numbers being shipped, the transportation mode, and the storage node when storage occurs.

As can be seen from Table 14, when the nominal transport prices in Γ_{rn} fall mostly during the price decline period in Table 4, that is, Γ_{rn} = (4,1,0), column 6 shows that order 3 is transported in period 3; order 111 is transported in periods 2 and 3, while Γ_n is constant and Γ_{rn} = (0,1,4), column 11 shows that order 3 occurs in period 1, and order 111

Table 10
Comparison between CPLEX and two heuristics.

		CPLEX		MKIGALNS			MGA		
Example	N	Optimal solution	Time(s)	Average solution	Time(s)	Gap%	Average solution	Time(s)	Gap%
S-n40	40	404 660	68.12	404 660	<1	0.00	419 720	<1	3.72
S-n48	48	492 370	82.56	492 370	<1	0.00	507 370	<1	3.05
S-n56	56	555 990	101.11	555 990	<1	0.00	571 510	<1	2.79
S-n64	64	634 000	115.34	634 000	<1	0.00	665 020	<1	4.89
S-n72	72	707 460	139.55	707 460	<1	0.00	729 770	<1	3.15
S-n80	80	771 700	144.23	771 700	<1	0.00	804 180	<1	4.21
S-n88	88	845 550	158.33	845 550	1.18	0.00	901 040	<1	6.56
S-n96	96	919 800	180.22	919 800	1.04	0.00	965 220	<1	4.94
S-n104	104	993 100	195.44	993 100	1.47	0.00	1 077 100	<1	8.46
S-n112	112	1 083 200	232.55	1 083 200	1.33	0.00	1 195 200	<1	10.34
Avg.		740 783	141.75	740 783	<1	0.00	783 613	<1	5.21

Table 11
Impact of in-transit storage on operating costs.

Number	Example	Control group		Experimental group		Gap
		Optimal solution	Time(s)	Optimal solution	Time(s)	%
1	S-n40	405 270	67	404 660	68.12	−0.15
2	S-n48	493 090	83	492 370	82.56	−0.15
3	S-n56	556 840	97	555 990	101.11	−0.15
4	S-n64	643 680	118	634 000	115.34	−1.53
5	S-n72	717 140	128	707 460	139.55	−1.37
6	S-n80	781 380	145	771 700	144.23	−1.25
7	S-n88	855 230	160	845 550	158.33	−1.14
8	S-n96	929 680	172	919 800	180.22	−1.07
9	S-n104	1 003 400	198	993 100	195.44	−1.04
10	S-n112	1 094 100	220	1 083 200	232.55	−1.01
Avg.		747 981	138.8	740 783	141.75	−0.89

Table 12
Impact of transit costs on operating cost.

Example	Scenario1			Scenario2			Scenario3		
	EG	CG	Gap(%)	EG	CG	Gap(%)	EG	CG	Gap(%)
S-n40	404660	405270	−0.15074	364730	364730	0	410230	420450	−2.49129
S-n48	492370	493090	−0.14623	443910	443910	0	504470	517070	−2.49767
S-n56	555990	556840	−0.15288	501340	501340	0	573720	587220	−2.35306
S-n64	634000	643680	−1.52681	576770	582710	−1.02987	651730	679660	−4.28552
S-n72	707460	717140	−1.36828	643740	649690	−0.92429	733980	763120	−3.97014
S-n80	771700	781380	−1.25437	705300	711240	−0.84219	801420	830560	−3.63605
S-n88	845550	855230	−1.14482	771600	777540	−0.76983	875270	904410	−3.32926
S-n96	919800	929680	−1.07415	839400	845340	−0.70765	953230	984360	−3.26574
Avg.	666441.3	672788.8	−0.85229	605848.8	609562.5	−0.53423	688006.3	710856.3	−3.22859

EG:Experimental group CG:Control group.

Table 13
Variation of transport strategy for different Γ_k.

Γ_k =112					Γ_k =72					Γ_k =40				
Periods	Routes	Orders	Modes	SN	Periods	Routes	Orders	Modes	SN	Periods	Routes	Orders	Modes	SN
k = 3	23-22	102	Railway		k = 3	24-21	106	Railway		k = 3	24-21	106	Railway	
	22-20	102&93	Railway	20		21-20	106	Waterway			21-20	106	Railway	20
k = 5	24-21	106	Railway			20-18	106	Waterway		k = 5	20-18	106&102&93	Waterway	
	21-20	106	Waterway			18-7	106	Waterway			18-7	106&102&93	Waterway	
	20-18	106&102&93	Waterway			7-8	106	Waterway			7-8	106&102&93	Waterway	
	18-7	106&102&93	Waterway			23-22	102	Railway			23-22	102	Railway	
	7-8	106&102&93	Waterway			22-20	102&93	Railway	20		22-20	102&93	Railway	
	8-6	93	Waterway		k = 5	20-18	102&93	Waterway			8-6	93	Waterway	
	6-2	93	Waterway			18-7	102&93	Waterway			6-2	93	Waterway	
						7-8	102&93	Waterway						
						8-6	93	Waterway						
						6-2	93	Waterway						

SN:Storage nodes.

Table 14
Variation of transport strategy for different (Γ_n , Γ_{rn})combinations.

Γ_n = 2, Γ_{rn} = (0,1,4)					Γ_n = 4, Γ_{rn} = (4,1,0)					Γ_n = 4, Γ_{rn} = (0,1,4)				
Periods	Routes	Orders	Modes	SN	Periods	Routes	Orders	Modes	SN	Periods	Routes	Orders	Modes	SN
k = 1	13–24	112	Railway		k = 1	13–24	112	Railway		k = 1	13–24	3 &112	Railway	
	24–21	112	Railway			24–21	112	Railway			24–21	3 &112	Railway	
	21–20	112	Waterway			21–20	112	Waterway			21–20	3 &112	Waterway	
	20–18	112	Waterway			20–18	112	Waterway			20–18	3 &112	Waterway	
	18–7	112	Waterway			18–7	112	Waterway			18–7	3 &112	Waterway	
k = 4	13–24	111	Railway		k = 2	13–24	111	Railway		k = 4	13–24	111	Railway	
	24–21	111	Railway	21		24–21	111	Railway	21		24–21	111	Railway	21
k = 5	13–24	3	Railway		k = 3	13–24	3	Railway		k = 5	21–20	111	Waterway	
	24–21	3	Railway			24–21	3	Railway			20–18	111	Waterway	
	21–20	3&111	Waterway			21–20	3&111	Waterway			18–7	111	Waterway	
	20–18	3&111	Waterway			20–18	3&111	Waterway						
	18–7	3&111	Waterway			18–7	3&111	Waterway						

SN:Storage nodes.

is transported in periods 4 and 5. When and Γ_{rn} = (0,1,4) is unchanged and Γ_n = 2, in column 1, order 3 is transported in period 5. Thus, when the combination of (Γ_n, Γ_{rn}) is changed, the order transportation time changes.

The amount of commodities transported by each road section in columns 8 and 13 of the table shows that when Γ_n = 4 is constant when Γ_{rn} = (4, 1, 0), road sections (20–18) use commodities from waterway transport order 112 in period 1 and commodities from waterway transport orders 3 and 111 in period 3. When Γ_{rn} = (0, 1, 4), road section (20–18) adopts the commodities of waterway transport orders 3 and 112 in period 1 and the commodities of waterway transport order 111 in period 5. When Γ_{rn} = (0, 1, 4) is constant, when Γ_n = 2, road segment (20–18) adopts goods of waterway transport order 112 in period 1 and goods of waterway transport order 3, 111 in period 5. Thus, when the combination of (Γ_n, Γ_{rn}) changes, the number of commodities transported by each road segment changes.

From the amount of goods stored in Warehouse 21 contained in columns 3, 8, and 13 of the table, it can be seen that when Γ_n = 4 and Γ_{rn} = (4, 1, 0), Warehouse 21 stores the amount of goods for order 111 at moment 2; when Γ_n = 4 and Γ_{rn} = (0, 1, 4), Warehouse 21 stores the amount of goods for order 111 at moment 4. Therefore, when the combination of (Γ_n, Γ_{rn}) changes, there is a change in the number of goods stored in each warehouse. Therefore, when Γ_{rn} changes, the period of transportation, the volume of commodities transported, and the period of storage, all change accordingly to reduce the negative impact of high freight rates on operating costs. When Γ_n increases, the period of transportation, the volume of commodities transported, and the period of storage, all produce corresponding changes, to reduce the negative impact of price fluctuations on operating costs.

This section further analyzes the trend of operating costs under 42 combinations of (Γ_n, Γ_{rn}), involving 6 Γ_n values and 7 Γ_{rn} values. As can be seen from Fig. 6, an increase in Γ_n for a certain Γ_{rn} will lead to higher operating costs, i.e., the more freight rate fluctuations occur for each mode of transportation, will lead to higher total costs. Meanwhile, for a given parameter Γ_n, all six curves in Fig. 6 show a decreasing trend, as the expected freight rate falls in the price decline period more often, i.e., the operating cost keeps decreasing. Since different decision makers have different preferences for risk, the optimal operating costs under different risk can be obtained by adjusting the values of uncertain budgets, to support the final planning of decision makers with different risk preferences.

6.5.5. Analysis of the impact of uncertain budgets on storage strategies

This section analyzes the effect of parameters(Γ_n, Γ_{rn}) on the storage strategy, so that Γ_n is kept constant and Γ_{rn} is taken as an example for three different values. Fig. 7 shows the storage of commodities in the transit warehouse. When Γ_n = 2 is kept constant, the location selection strategies of the transit warehouse are similar in the three cases of Γ_{rn} = (0, 1, 4), Γ_{rn} = (2, 1, 2), and Γ_{rn} = (4, 1, 0). For example, the transit warehouse numbered {3, 4, 5,6, 8, 9,11, 13,17, 19, 20, 21,22,24} are selected in all three different Γ_{rn}. The transit warehouse numbered 16 is used only when Γ_{rn} = (2, 1, 2). At the same time, different combinations of parameters (Γ_n, Γ_{rn}) have a significant effect on the storage volume of the commodities in the transit warehouse, with only a few transit warehouses not being significantly affected, such as the transit warehouse numbered {17, 21, 24}.

7. Case study

Company Z is a multifunctional intelligent logistics platform company in Zhengzhou City that mainly conducts domestic and international multimodal transport, import and export logistics, and other businesses. In this section, we build a series of reality-based case studies based on the research data of multimodal transport company Z (specifically, the data include transport network, transport price, commodity type, and order quantity) and process them using the MKIGALNS algorithm. The relevant parameters and descriptions are as follows.

7.1. Description of instances

(1) The Size of instances: To generate examples in a practical context, this study selects the current routes of Company Z in the above dataset (starting from Zhengzhou and transporting to countries such as Russia, Japan, and South Korea via Qingdao harbor, Lianyun harbor, and Beilun harbor) and combines them with the expansion and extension to future markets(i.e., the development of the urban area around Zhengzhou as a multimodal starting point and the collection and transport of cargoes from various counties to the urban area) to form a multimodal transport network for the case study in this section, as shown in Fig. 8. As shown in Fig. 8, the multimodal transport network consists of 17 nodes, including China, Japan, South Korea, and Russia. The red dots represent candidate nodes for commodity storage and transportation-mode conversion. The yellow connecting lines represent the transportation routes.

(2) Real data are used: this section investigates the data from company Z and obtains the nodes of this experiment in conjunction with future market expansion. The waterway transport distance between the nodes is obtained through a map mapping website. The highway and railway distances between the nodes are obtained through a geographic information website (owing to space constraints, only some of the distances are shown in Table 15, where the symbol "*" indicates that the mode of transport does not exist). Through research and consulting the relevant websites of the state and transportation service companies, we can get information on freight rates in the past few years, which fluctuate greatly by waterway, reaching an average

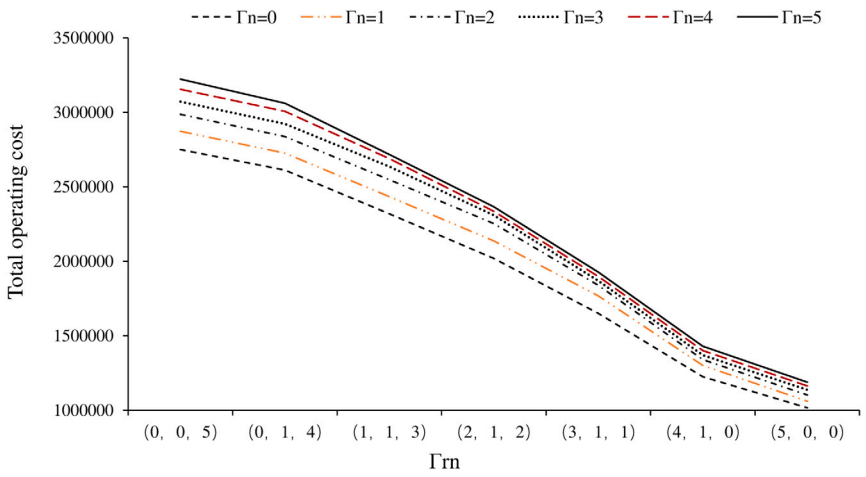


Fig. 6. Impact of freight rates uncertainty budget(Γ_n , Γ_{rn}) on operating costs.

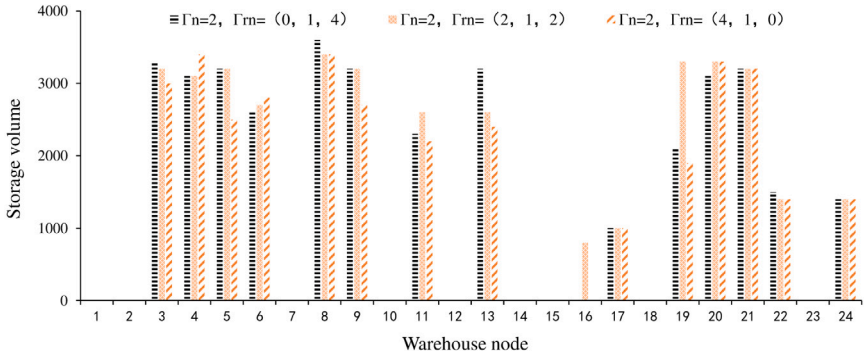


Fig. 7. Effect of different (Γ_n, Γ_{rn}) on storage warehouses and storage volume.

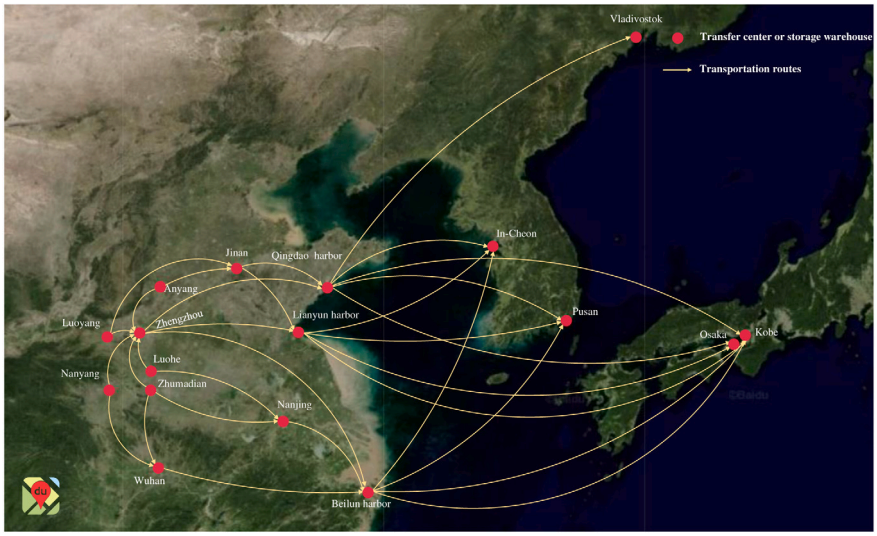


Fig. 8. Map of multimodal transport routes.

of 70%–80%, with some routes experiencing a range as high as 200%–300%; The railway has a price range of 30%, and the highway has a price range of 50%. Therefore, based on data from research and related websites, we obtained nominal prices for each price range based on 80% price fluctuations for waterway transport, 30% price fluctuations for railways, and 50% price fluctuations for highway, as listed in Table 16. This study assumes that the order q = 80 and the period k = 5.

7.2. Impact of in transit storage on operating costs

Based on the background of the real case, this section assumes that the transport price and storage space are determined. By setting up

Table 15
The order route and the associated distance.

Order route	HD	RD	WD	Order route	HD	RD	WD
Nanyang-Zhengzhou	279	369	*	Luoyang-Zhengzhou	156	124	*
Zhengzhou-Beilun harbor	1101	1129	*	Zhengzhou-Qingdao harbor	721	1042	*
Beilun harbor-Kobe	*	*	1362	Qingdao harbor -Osaka	*	*	1387
Zhumadian-Zhengzhou	213	206	*	Luohe-Nanjing	532	670	*
Zhengzhou-Lianyun harbor	553	539	*	Nanjing-Beilun harbor	458	642	*
Lianyun harbor-Pusan	*	*	895	Beilun harbor-In Cheon	*	*	963

HD: Highway distance RD: Railway distance WD: Waterway distance.

Table 16
nominal freight rate for different rate ranges.

freight rate (Yuan/ton)	Highway	Railway	Waterway
Price reduction period	0.334	0.096	0.0019
Price normal period	0.500	0.120	0.040
Price increase period	0.667	0.144	0.062

a control experimental group, the experimental results of the control group are compared and analyzed with the multimodal transport path and storage co-optimization strategy proposed in this study with the same parameter values. The control group was not allowed to store during the transport tasks. The experimental results are listed in Table 17. Columns 2–3 were divided into the two experimental groups' total operating and storage costs. Column 4 shows the randomly selected orders used to observe whether there is a difference in transport strategies. Columns 5–7 show the periods in which orders are transported, the transport routes in each period, and the nodes at which storage occurs.

The values in Table 17 show that when storage does not occur, the control group transports commodities of order 78 from Nanyang to the endpoint (Kobe) in period 3. When storage is allowed, the experimental group first transports commodities of order 78 from Nanyang to the storage warehouse (Beilun harbor) in period 3, and after storing them for one period, transports the commodities in the warehouse to the end point (Kobe) in period 5. Therefore, storage affects the transport strategy of a commodity. From column 2, when storage does not occur, the total operating cost of the control group is 773270; when storage is allowed, the total operating cost decreases to 670310, which is a 13.31% decrease in the total operating cost. When storage was allowed, the experimental group had a storage cost of 27240, accounting for 4.06% of the total operating costs. Thus, it is evident that the strategic storage of some items helps reduce the overall operating costs. In multimodal transport, managers can establish warehouses at key locations, resulting in more flexible transport strategies and lower costs.

7.3. Impact of transport price uncertainty on transport strategies

This section analyzes the impact of changes in the magnitude of price fluctuations in different transport modes on multimodal transport strategies when transport prices are uncertain. In the experiments described in this section, it is assumed that $\Gamma_n = 3$, $\Gamma_{rn} = (2,1,2)$, and the maximum deviation of the transport price from its nominal value is 20%, that is, $\hat{c}^r_{nk} = 0.2\bar{c}^r_{nk}$. It is demonstrated that when the nominal price of the different modes of transport in different ranges of price intervals in the robust model is changed, there are changes in the total operating cost, transshipment cost, storage cost, transport time, and storage time. The experimental results are listed in Table 18. In particular, column 1 indicates the three price fluctuation range scenarios used in this section,where Scenario 1 indicates that waterway price fluctuation is 80%, railway price fluctuation is 30%, and highway price fluctuation is 50%; Scenario 2 indicates that waterway price fluctuation is 200%, railway price fluctuation is 30%, and highway price fluctuation is 50%; and Scenario 3 indicates that waterway price fluctuation is 200%, railway price and highway price fluctuation is 0. Columns 2–3 show the total operating and storage costs for each of the three price fluctuation scenarios. Column 4 contains randomly selected orders used to determine whether there are differences in transportation strategies. Columns 5–7 show the periods in which the orders are transported, transport routes in each period, and nodes in which storage occurs.

As can be seen from the values in Table 18, the transport route for the goods of order 78 in all three cases was to transport the storage warehouse (Beilun harbor) from Nanyang, store the goods for one period, and then transport the goods in the warehouse to the final destination (Kobe). However, the transportation and storage times of the commodities, in the order of 78, changed in all three scenarios. Under the price fluctuation range of scenario 1, the transportation of the goods of order 78 occurs in periods 2 and 3, and the storage decision occurs in the 2nd period; under the price fluctuation range of scenario 2, the transportation of the goods of order 78 occurs in periods 3 and 4, and the storage decision occurs in the 3rd period; under the price fluctuation range of scenario 3, the transport of goods of order 78 occurs, and the storage decision occurs in period 1; therefore, the magnitude of price volatility of different transport modes has an impact on the transport strategy of the commodity.

And from column 2, the total operating cost of scenario 1 is 1085100, of which the storage cost is 10320, accounting for 0.95% of the total operating cost; the total operating cost of scenario 2 is 967330, of which the storage cost is 10680, accounting for 1.10% of the total operating cost; the total operating cost of scenario 3 is 948770, of which the storage cost is 10680, accounting for 1.13% of the total operating cost; thus, it can be seen that price fluctuations of different modes of transport have an impact on the overall operating cost. Due to the volatility of transport prices, managers can use robust modeling to make trade-offs between total operating costs and timeliness in meeting transport demands, giving them a satisfactory transport strategy.

8. Management implications

This study investigates the problem of considering multi-period multimodal transport routes and storage strategies in uncertain environments. When building the model, in addition to the transportation cost of multitype transportation, transfer and storage costs were included in the total operating cost function. Faced with the uncertain transportation price in the multi-type transportation process and the storage space of the product, this study uses polygonal uncertainty and the robust optimization model for processing. Subsequently, a series of algorithm analysis experiments at different scales were conducted to verify the model's accuracy and effectiveness of the MKIGALNS algorithm solution. In addition, this study analyzes an instance based on a realistic background to enhance its reliability. The results of this study have practical significance, as they can help enterprises optimize resources and better adapt to the fluctuation of freight rates to minimize operating costs. The model recommended in this study has the following management implications.

First, it describes the impact of storage on multi-type transport operating costs. To a certain extent, storage decision-making affects the flexibility of choosing a transportation mode. When K is at a time when there is a capacity constraint or a high transportation price, the goods are stored at the correct storage point according to the characteristics

Table 17
Impact of in-transit storage on operating costs.

	TC	Storage cost	Order	Periods	Routes	Storage node
Control group	773270	0	78	k=3	Nanyang–Zhengzhou—Beilun harbor===Kobe	
Experimental group	670310	27240	78	k=3 k=5	Nanyang–Zhengzhou–Beilun harbor Beilun harbor===Kobe	Beilun harbor

TC: Total operating cost —Railway ===Waterway.

Table 18
Impact of freight rate uncertainty on transport strategies.

	TC	Storage cost	Order	Periods	Routes	Storage node
Scenario1	1085100	10320	78	k=2 k=3	Nanyang—Zhengzhou—Beilun harbor Beilun harbor===Kobe	Beilun harbor
Scenario2	967330	10680	78	k=3 k=4	Nanyang—Zhengzhou—Beilun harbor Beilun harbor===Kobe	Beilun harbor
Scenario3	948770	13020	78	k=1 k=2	Nanyang—Zhengzhou—Beilun harbor Beilun harbor===Kobe	Beilun harbor

TC: Total operating cost —Railway ===Waterway.

and needs of the goods. This flexibility enables enterprises to choose economical and efficient transportation methods in different situations and reduces transportation costs. The storage point can be reasonably set to help reduce the transit link and directly affect the overall transportation costs. By setting up temporary storage facilities at critical positions, seamless connections of different transportation methods can be achieved, reducing the loss and additional handling costs of cargo in the transit process and the overall reduction of transfer and storage costs. As shown in Section 6.5.1, in the ten test cases, the optimal operating costs of the control group (the control group was not allowed to store commodities during transportation) were greater than those of the experimental group on average. As shown in Section 6.5.2, transit costs have a corresponding impact on operating costs, and storage and transshipment coordinate with each other, jointly influencing operating costs. As shown in Section 7.2, when storage does not occur, the total operating cost of the control group is 773270; when storage is allowed, the total operating cost decreases to 670310, which is a 13.31% decrease in the total operating cost. When storage was allowed, the experimental group had a storage cost of 27240, accounting for 4.06% of the total operating costs.

Secondly, this study not only considers the uncertainty of the transport price and the uncertainty of the storage space of the commodity but also uses the polyhedral uncertainty set robust optimization method, which provides a new perspective to improve the robustness of logistics and transport and reduce risk. Under this model, enterprises can adjust the impact of different price scenarios or different types of product types on the multi-transport transportation system by adjusting the size of the uncertain budget. The formula transportation strategy helps enterprises adapt better to fluctuations in freight rates and changes in the types of transportation products, reduces transportation costs, and improves competitiveness. As shown in Section 6.5.4, when robust parameters with uncertain storage space changes or robust parameters with uncertain freight rate changes, the period of transportation, volume of commodities transported, and volume of storage all produce corresponding changes to reduce the negative impact of transport commodities of different sizes and price fluctuations on operating costs. Section 6.5.5 shows the changes in the warehouse point where storage occurs and the warehouse's storage capacity when the uncertain budget changes.

Finally, this study considered the impact of the amplitude of different prices on transportation strategies. In Section 7.3, when price fluctuations change, orders' transportation and storage times also change. Therefore, in an uncertain multitype joint transport market environment, this model can help enterprises make full use of the differences in price fluctuations in different periods through reasonable arrangements and storage periods and flexibly adjust transportation and storage strategies to better realize efficient, flexible, and sustainable logistics transportation.

9. Conclusion

This study investigated the problem of optimizing multitemporal and multimodal transport routes and transit, by considering storage service strategies. First, an integer-programming mathematical model, is used to describe the problem in a deterministic setting. Combined with multimodal operation practices, the storage costs incurred in the multimodal transport process are incorporated into the objective function. Commodities can go to the transit warehouse for storage in transit, and commodities can be transported to the transit warehouse or terminal in advance, which intertemporal transport are allowed. Second, this study explores the synergistic optimization of multitemporal multimodal transport routes and storage strategies, under freight rate and storage space uncertainty. To more accurately model the uncertainty of freight rates in cross-border trade, and to avoid overly conservative results, this study adopts a robust optimization method with uncertain parameters characterized by multiple ranges, and enhances its solution efficiency through robust pairwise equation transformation. We propose a hybrid heuristic algorithm MKIGALNS combining the traditional k-shortest path algorithm, the iterative greedy algorithm (IG) and the adaptive large neighborhood search algorithm (ALNS) to solve the problem, and then verify the accuracy of the model and the effectiveness of the algorithm through a series of small-scale arithmetic case experiments. Finally, numerical experiments were conducted using arithmetic examples of different scales and a series of case experiments based on research data to provide management insights for enterprises.

In a series of small-scale experiments, we compared the performance of CPLEX, MGA algorithm and MKIGALNS algorithm. The results show that the MKIGALNS algorithm outperforms the CPLEX and MGA algorithms in terms of overall solution efficiency. Multiple numerical experiments of different sizes were conducted in this study. The experimental results show that strategic storage and transit of some commodities can help reduce the overall operating costs, in the presence of large variations in freight rates and storage space requirement, and severe capacity constraints. In addition, the solution of the robust model revealed the significant impact of freight rate and storage space uncertainty on the multi-period intermodal operation strategy. Including multi-period multimodal operation strategies that allow order splitting, which the present study did not consider, and designing efficient algorithms, will be the focus of future work.

CRediT authorship contribution statement

Fang Guo: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Funding acquisition, Conceptualization. **Yan Xu:** Writing – review & editing, Writing – original draft, Software. **Zhihong Huang:** Writing – review & editing, Writing – original draft, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Yunxiang Wu:** Writing – review & editing, Writing – original draft, Software, Methodology.

Data availability

Data will be made available on request.

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